

**KADIR HAS UNIVERSITY**  
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**Silent Speech Interface (SSI):**  
**A Wearable Multimodal Biosignal System for Post-Laryngectomy Voice Restoration**

**Group Members:**

Tuğrul Altınüzengi

Serhat Yalçın

Mert Kaçmaz

Yiğit Efe Ayhan

**Supervisor: Yalçın Şadi**

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## ABSTRACT

Post-laryngectomy patients face the permanent loss of biological voice production, creating significant barriers to social participation and quality of life. This report presents the design and preliminary evaluation of a wearable Silent Speech Interface (SSI) that restores non-acoustic communication through multimodal biosignal sensing. The device integrates one surface Electromyographic (EMG) channel and four Piezoelectric (PZT) transducers mounted in a 3D-printed cervical band, capturing neuromuscular activation patterns and mechanomyographic tissue vibrations from the submental and lateral cervical regions.

A real-time digital signal processing pipeline — implemented on an Android mobile application — performs causal Butterworth bandpass filtering (EMG: 20–200 Hz; PZT: 5–200 Hz), 50 Hz mains rejection via a high-Q IIR notch filter, and exponential moving average envelope extraction for voice activity detection. Validated against manually annotated speech markers, the VAD system achieved a 66% word-level match rate at an optimised threshold multiplier of  $k = 1.5$ , across ten target Turkish vocabulary items in a 65-second pilot recording session.

A one-dimensional Convolutional Neural Network (1D-CNN) operating on raw ( $5 \times 1000$ ) sensor tensors targets classification of ten Turkish words. A Random Forest baseline achieved 67% within-session top-1 accuracy and 87% top-3 accuracy on hand-crafted 61-element feature vectors, confirming the presence of class-separable articulatory patterns in the collected data. The full CNN pipeline, including ONNX model export and on-device Android inference, is designed for sub-200 ms end-to-end latency to preserve natural conversational flow.

The system's design framework addresses international regulatory standards (IEC 60601-1, ISO 10993, TİTCK/MDR 2017/745), environmental sustainability requirements (EU RoHS Directive PZT exemptions), KVKK/GDPR biometric data protection mandates, and three United Nations Sustainable Development Goals (SDGs 3, 9, and 10). Ongoing work focuses on multi-session dataset expansion, cross-session model generalisation, and clinical validation with the target patient population.

# Table of Contents

|                                                                                |    |
|--------------------------------------------------------------------------------|----|
| ABSTRACT .....                                                                 | 2  |
| 1 INTRODUCTION.....                                                            | 6  |
| 2 LITERATURE REVIEW .....                                                      | 8  |
| 3 METHODOLOGY .....                                                            | 11 |
| 3.1 Comparative Analysis of Existing Voicing Solutions .....                   | 11 |
| 3.2 Proposed Silent Speech Interface Methodology and Sensor Architecture ..... | 12 |
| 3.2.1 Anatomical Sensor Placement and Rationale .....                          | 13 |
| 3.2.2 Sensor Fusion and Decoding Algorithm .....                               | 14 |
| 3.3 Signal Conditioning and Pre-processing .....                               | 15 |
| 3.3.1 Frequency Filtering and Artifact Mitigation .....                        | 15 |
| 3.4 Voice Activity Detection (VAD) and Segmentation .....                      | 16 |
| 3.4.1 Envelope Calculation and Dynamic Thresholding.....                       | 16 |
| 3.4.2 State Machine Logic and Pre-roll Buffering .....                         | 17 |
| 3.5 Dataset Construction and Feature Extraction .....                          | 17 |
| 3.5.1 Automated Data Pipeline and Label Mapping .....                          | 17 |
| 3.5.2 Feature Extraction Methodology .....                                     | 20 |
| 3.6 Machine Learning Architectures and Preliminary Evaluation .....            | 21 |
| 3.6.1 Baseline Classification (Random Forest) .....                            | 21 |
| 3.6.2 Preliminary Results (Legacy Data) .....                                  | 21 |
| 3.6.3 Deep Learning Framework (1D-CNN) .....                                   | 22 |
| 3.6.4 Data Augmentation Strategies .....                                       | 22 |
| 3.7 Hardware Adaptation and System Revisions .....                             | 23 |
| 3.7.1 Transition to 5-Channel Architecture .....                               | 23 |
| 3.7.2 Modality Ablation and Sensor Impact Studies .....                        | 24 |
| 3.8 Edge Deployment and Cross-Platform Integration .....                       | 24 |
| 3.8.1 Neural Network Serialization (ONNX).....                                 | 25 |
| 3.8.2 Digital Signal Processing (DSP) Parameter Synchronization.....           | 25 |
| 4 CONSTRAINTS, CONDITIONS, AND REQUIREMENTS.....                               | 26 |
| 4.1 Anatomical and Design Constraints .....                                    | 26 |
| 4.2 Signal Acquisition Conditions.....                                         | 26 |
| 4.3 Computational and Operational Requirements .....                           | 27 |
| 5 PROJECT CALENDAR AND PROJECT BUDGET .....                                    | 29 |
| 5.1 Project Calendar .....                                                     | 29 |
| 5.2 Project Budget .....                                                       | 29 |

|                                                                                       |    |
|---------------------------------------------------------------------------------------|----|
| 6 SYSTEM DESIGN AND ANALYSIS .....                                                    | 31 |
| 6.1 Hardware Architecture .....                                                       | 31 |
| 6.1.1 Power and Energy Management.....                                                | 31 |
| 6.1.2 Data Acquisition (DAQ) and Wireless Node .....                                  | 31 |
| 6.2 Mechanical Design .....                                                           | 31 |
| 6.2.1 Ultralight PLA Housing .....                                                    | 31 |
| 6.2.2 Sensor Attachment and Coupling .....                                            | 32 |
| 7 IMPLEMENTATION PROCESS.....                                                         | 60 |
| 7.2 Mechanical Implementation: 3D Scanning and Custom Collar Fabrication.....         | 60 |
| 7.3 Data Acquisition Protocol for Machine Learning .....                              | 63 |
| 7.4 Embedded Architecture and Wireless Mobile Integration .....                       | 64 |
| 7.5 Personalized Text-to-Speech (TTS) Voice Cloning Integration .....                 | 65 |
| 8 PRODUCT TESTING AND SYSTEM VALIDATION PLAN .....                                    | 66 |
| 8.1 Algorithm Performance and Decoding Reliability .....                              | 66 |
| 8.1.1 Phase 0: Baseline Verification Results (Random Forest).....                     | 66 |
| 8.1.2 Phase 1: Advanced Target Validation (1D-CNN).....                               | 67 |
| 8.2 Voice Synthesis Personalization and Acoustic Fidelity.....                        | 68 |
| 8.2.1 Acoustic Profile Extraction and Voice Cloning.....                              | 68 |
| 8.2.2 Perceptual Validation and 90% Similarity Benchmark .....                        | 68 |
| 8.3 Clinical Validation Protocols and Multi-Phasic Testing.....                       | 68 |
| 8.3.1 Phase I: Pre-Clinical Validation with Healthy Control Group .....               | 69 |
| 8.3.2 Phase II: Clinical Trials with Total Laryngectomy Patients.....                 | 69 |
| 8.4 Real-Time System Performance and Computational Resource Profiling.....            | 70 |
| 8.4.1 End-to-End Latency Budget and Temporal Validation .....                         | 70 |
| 8.4.2 Hardware Resource Consumption and Energy Efficiency .....                       | 70 |
| 8.5 Data Integrity, Privacy Validation, and Biometric Security (KVKK Compliance)..... | 71 |
| 8.5.1 Biometric Encryption and Localized Inference Validation .....                   | 71 |
| 8.5.2 Data Encoding and Integrity Verification.....                                   | 71 |
| 8.6 Technical Documentation and Software Life Cycle Validation (IEC 62304) .....      | 71 |
| 8.6.1 Unit Testing and Regression Analysis .....                                      | 71 |
| 8.6.2 Vulnerability Assessment of the ML Pipeline .....                               | 72 |
| 9 STANDARDS .....                                                                     | 72 |
| 9.1 Electrical and Electronic (EE) Engineering Standards .....                        | 72 |
| 9.2 Biocompatibility Standards .....                                                  | 73 |
| 10 SUSTAINABLE DEVELOPMENT .....                                                      | 75 |

|                                                              |    |
|--------------------------------------------------------------|----|
| 10.1 Goal 3: Good Health and Well-Being .....                | 75 |
| 10.2 Goal 9: Industry, Innovation, and Infrastructure .....  | 76 |
| 10.3 Goal 10: Reduced Inequalities .....                     | 76 |
| 11 IMPACTS .....                                             | 77 |
| 11.1 Environmental Impacts.....                              | 77 |
| 11.2 Health Impacts.....                                     | 78 |
| 11.3 Legal Consequences .....                                | 79 |
| 12 EXPERIMENTAL RESULTS AND FINDINGS.....                    | 80 |
| 12.1 Signal Acquisition Protocol and Session Timing .....    | 80 |
| 12.2 System Performance Evolution .....                      | 83 |
| 12.3 Word-Level Cross-Session Classification Accuracy .....  | 84 |
| 9 CONCLUSION .....                                           | 86 |
| References .....                                             | 87 |
| APPENDIX .....                                               | 94 |
| A.1 Transport Layer and Data Acquisition.....                | 94 |
| A.1.1 Bit-Level Framing Protocol.....                        | 94 |
| A.2 Real-Time Digital Signal Processing (DSP) .....          | 95 |
| A.2.1 Mathematical Foundations of the Filter Topologies..... | 95 |
| A.2.2 Two-Stage Preprocessing Pipeline .....                 | 95 |
| A.3 Initialization and Calibration Algorithms.....           | 96 |
| A.4 Voice Activity Detection (VAD) Segregation.....          | 96 |
| A.5 Machine Learning Inference Pipeline .....                | 96 |

# 1 INTRODUCTION

A total laryngectomy (TL) is an extreme change in lifestyle operation, and is indicated primarily for treatment of advanced laryngeal or hypopharyngeal cancer, severe radiation damage of the larynx, or irreparable laryngeal injuries. For this operation, the larynx (including its vocal folds), thyroid cartilage, and cricoid cartilage will be completely removed from the neck in one piece (en bloc). The trachea will also be surgically redirected and connected to the front of the neck, and a tracheostoma (permanent stoma used for breathing) will be created in the neck [1] Although this is often necessary to save one's life and treat cancer, it drastically disrupts the connection of the larynx with the rest of the human aerodigestive tract, resulting in the complete separation of the entrance to the airway from the structures that make up the use of those sounds via the mouth [2]

The most significant functional loss caused by a total laryngectomy is the complete lack of biological phonation. In a normally functioning human being, speech is produced by the aerodynamic-myoelastic model of phonation, whereby the pulmonary airflow is forced through the adducted vocal folds and thus producing a vibrating acoustic source that is filtered and shaped by the movements of the tongue, jaw and lips [4] After total laryngectomy, the pulmonary air is prevented from reaching the articulators. As a result of the permanent alaryngeal nature of the patient, the impact of the loss of biological voice results in profound negative implications on the patient. The sudden loss of the ability to spontaneously and effortlessly communicate verbally creates an enormous amount of psychological stress, creates significant social withdrawal, affects the patient's ability to earn a living, and often leads to major depression [3]

The primary goal for restoring functional, intelligible, and natural-sounding communication after surgery to the head and neck is post-operative rehabilitation of the patient. For many years, three common methods of producing alaryngeal voices have been used by clinicians: pneumatic/electronic artificial larynx (electrolarynx), esophageal speech (ES), and tracheoesophageal puncture (TEP) using a one-way silicone voice prosthesis. .2 Although all three methods have been the foundation for speech rehabilitation for many years, they each have substantial limitations related to their physical, physiological, acoustic, and socioeconomic nature [1]

There is still no complete or seamless solution for individuals who have had their voice box surgically removed; this is demonstrated by the limitations of using an electrolarynx, the difficulties associated with using esophageal speech, and the continuous surgical, infection-related, and financial difficulties associated with the use of a TEP prosthesis [7] These ongoing clinical problems, as well as the acoustic shortcomings of esophageal speech and the issues of sound generated by a TEP, illustrate the critical need for a significant technological advancement. New Silent Speech Interfaces (SSIs) represent a significant development that may be an entirely new form of communication for laryngectomized individuals that requires no surgical procedures. The SSI captures recorded neuromuscular and mechanomyographic (EMG and MMG) data indicating what sound the user wants to speak and translates that data into synthesized speech, without using the surgically distorted vocal tract [10] This detailed research report explains how this new SSI uses both surface electromyography and piezoelectric lead zirconate titanate (PZT) sensors to give back to someone who has had their voice box surgically removed the ability to communicate independently.

## 2 LITERATURE REVIEW

There has been a long standing documented history in both clinical and biomedical literature of efforts to compensate for a missing larynx. The primary function of every alaryngeal speech option is to provide a replaceable sound source and to create speech using the pharynx, mouth and nose as the articulatory cavities.

The electrolarynx, developed early on for use outside of the human body, is the most common form of external device. The modern trans-cervical electrolarynx is a battery-operated handheld unit that employs mechanical vibrations. When placed firmly against the tissues of the submandibular (under the chin) or cervical (neck) regions, the electrolarynx emits a continuous, monotonal sound wave from which the user forms their spoken words. The primary benefit cited in the literature is the relative ease of obtaining one; patients can obtain basic functional speech almost immediately following their operation with minimal speech therapy. The sound output produced by the electrolarynx has been characterized in the literature as "robotic," "unnatural" and lacking any of the prosodic cues (pitch, intonation, etc.) associated with natural speech. In addition, the use of a handheld device requires one of the user's hands be free for use, thus providing visibility and emphasis to the user's disability in a social setting.

The oldest non-surgical method for using the voice without a larynx, or alaryngeal communication, is through the use of esophageal speech. The process of using esophageal speech (ES) is that the patient blows air into the upper esophagus by purposely forcing air in through the mouth into the upper esophagus. When the patient forces out the air that they have blown into the upper esophagus, this creates vibrations in the upper pharynx and esophagus, which serve as the sound source for the voice box. Although the use of ES does not require any external support or maintenance after mastering it, very few individuals who have had a laryngectomy are able to completely adapt ES or use it consistently as their common method of communicating. Based on ongoing reviews of individuals who have had a laryngectomy, only about 30% of these individuals are able to use ES continually or as their primary means of communicating. The cognitive demand necessary to use ES including memory, learning how to coordinate the muscles involved, etc. can place a significant physical or emotional demand on a patient using ES and create muscle tension and even gastric symptoms.

Tracheoesophageal (TE) voice restoration is currently considered the worldwide clinical gold standard. Two methods for TE restoration exist:

1) Through tracheostomy placement, which will provide the ability to create a small fistula by surgically placing a prosthetic device in the esophagus, allowing for the production of phonation sound.

2) By blocking the trachea with a stoma (opening from trachea) and passing pulmonary air into the prosthesis via the valve and then directing that air into the esophagus to produce phonation sound.

TE restoration provides the most fluent, natural sounding alaryngeal voice (producing voice without the use of the larynx) with very long maximum phonation times because it uses the available robust flow of pulmonary air. However, many studies document the significant clinical issues associated with TE voice restoration.

TE prosthetics have a high incidence of *Candida albicans* biofilm colonization, which is the accumulation of yeast (*Candida albicans*, a fungus type that naturally live in the human body) that creates a protective layer. That can result in valve wear, leading to improper functioning of the voice prosthetics and the possibility of air leaking into the airway. As a result, there is an increased need for frequent replacement of prosthetics, resulting in excessive long-term costs to the patient, insurance carriers, and government payers.

In addition, periprosthetic leaks, stoma stenosis, and formation of granulation tissue on the prosthetic can cause the TE voice prosthetic to be ineffective or not work properly. And thus, TE voice prosthetic devices are very expensive to maintain and have numerous performance problems and clinical issues.

To address the limitations of compensatory mechanisms that rely on sound, researchers are now developing silent speech interfaces (SSIs). SSIs use technologies to interpret the intention to produce speech without detecting any sound during the articulation of words<sup>12</sup>. The literature describing various modalities used in SSIs has included electromagnetic articulography (EMA), which tracks the motion of the tongue relative to coils of a magnetic field; and diagnostic ultrasound (US) which provides an image of the vocal tract in real time [4] However, both methods have disadvantages including being large, requiring substantial computations, digital processing power and requiring intrusive use on a daily basis [4]

Surface Electromyography (sEMG) is an incredibly promising non-invasive technique [14] It detects the electrical signals, or action potentials, of muscle units as they contract. Specifically, the muscles in one's face and neck as they speak. This means that since the

pathways in the nervous system that control the tongue, lips, and jaw are undisturbed after a laryngectomy, sEMG is able to pick up the intention for speech from the brain [10] A typical sEMG system may decode the electrical signals from the muscles very accurately. However, they typically result in a low level of accuracy when identifying where the signals originate within the body as well as producing a lot of interference due to motion, which makes it difficult to distinguish between speech sounds and non-speech sounds [15] As such, the cutting-edge approach to SSI research is to fuse together the results from the sEMG with the results from mechanomyography transducers including Piezoelectric Lead Zirconate Titanate (PZT) sensors that convert and capture very small, low-frequency vibrations and variations of pressure from the deformations of the skin during speech [15] Using the results from both systems creates a synergy that eliminates the individual limitations of both technologies, resulting in unprecedented accuracy of decoding.

### 3 METHODOLOGY

#### 3.1 Comparative Analysis of Existing Voicing Solutions

To conduct a comprehensive methodology-based assessment of the suggested sensor-fusion SSI, it is necessary to first understand the specific characteristics, advantages and disadvantages; of each of the current clinical options. Table 3.1 contains an extensive comparative overview of the strengths and weaknesses of the electrolarynx, esophageal speech and tracheoesophageal puncture according to recent clinical evidence [2].

Table 3.1: Comprehensive methodological comparison of conventional alaryngeal voice restoration techniques.

| Voice Restoration Modality    | Mechanism of Action                                                                                                                                                                      | Clinical and Functional Advantages                                                                                                                                                                                                                                                         | Clinical and Functional Disadvantages                                                                                                                                                                                                                                                                                                 |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Electrolarynx (EL)</b>     | An electromechanical transducer is applied to the neck and produces a continuous vibrating frequency transmitted into the oral cavity for articulating speech.[2]                        | <ul style="list-style-type: none"><li>- Immediate post-surgery usability.</li><li>- Easy to learn, minimal or no speech therapy required.</li><li>- Functions efficiently in environments with high ambient noise level.</li><li>- No secondary surgical procedures required.[1]</li></ul> | <ul style="list-style-type: none"><li>- Distinctly robotic, mechanical type of sound.</li><li>- Monotonic sound prevents emotional accents and emphases.</li><li>- Loss of hand function during use, on the hand that holds the device.</li><li>- Device uses batteries for operation and requires periodic replacement.[2]</li></ul> |
| <b>Esophageal Speech (ES)</b> | Air from the ambient environment will be injected into the upper esophagus and will subsequently be expelled to vibrate the pharyngoesophageal (PE) segment to serve as a neoglottis.[1] | <ul style="list-style-type: none"><li>- Completely hands-free communication.</li><li>- No external electronic devices are used and therefore no batteries will be used.</li><li>- No surgical prostheses are needed, thereby eliminating ongoing maintenance.</li></ul>                    | <ul style="list-style-type: none"><li>- Extremely steep learning curve that goes up to 12 months with high failure rates of &gt;70%.</li><li>- Phonation is restricted to very short, quiet phrases.</li><li>- Produces considerable physical stress, tension in the muscles and</li></ul>                                            |

|                                         |                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         |                                                                                                                                                                           | - More natural sounding than the EL.[1]                                                                                                                                                                                                                                                                                                           | gastrointestinal issues such as bloating and flatulence.[1]                                                                                                                                                                                                                                                                                                                                    |
| <b>Tracheoesophageal Puncture (TEP)</b> | A one-way silicone prosthesis is surgically implanted into the esophagus and pulmonary air is shunted through the prosthesis to provide air to vibrate the PE segment.[6] | <ul style="list-style-type: none"> <li>- This is currently accepted as the gold standard for clinical use.</li> <li>- Utilizes pulmonary air, enabling loud, fluent speech with long maximum phonation times.</li> <li>- Has the most natural acoustic sound.</li> <li>- Can be used hands-free with some specialized stoma valves.[6]</li> </ul> | <ul style="list-style-type: none"> <li>- Requires primary or secondary surgical creation of a fistula.</li> <li>- Prosthesis is highly prone to fungal degradation, Candida albicans biofilm colonization.</li> <li>- High risk of periprosthetic leakage and aspiration.</li> <li>- Requires detailed daily hygiene, frequent clinical replacements and has high ongoing costs.[1]</li> </ul> |

According to the data in table 3.1, TEP provides excellent acoustic output; however, the surgical and maintenance burden associated with TEP makes it unattainable or intolerable for many of the patients who could benefit from it. The SSI design proposed here has been specifically designed to completely eliminate the physical and surgical limitations, while still maintaining the excellent intelligibility properties of fluent speech.

### 3.2 Proposed Silent Speech Interface Methodology and Sensor Architecture

The engineered solution is a highly specialized, non-invasive wearable band that captures the neuromuscular and mechanomyographic dynamics of the submental and cervical areas. The system uses an array of sensors consisting of one pair of Electromyography (EMG) sensors, four Piezoelectric Lead Zirconate Titanate (PZT) sensors, and one Reference (REF) electrode to provide robust open vocabulary speech decoding without the need for any type of acoustic vocalization. The calibrated placement of these sensors is essential to utilizing the decoding algorithm effectively as it defines where in space the musculature that corresponds to articulations is located.

### 3.2.1 Anatomical Sensor Placement and Rationale

When placing sensors on the skin, they are located at anatomical sites of the lower face and neck that were not extensively remodeled during or after total laryngectomy.

**PZT Sensor Array (MMG):** The use of PZT sensors allows for the detection of very small pressure changes and vibrations received through the epidermal layer as a result of motor activities (jaw, base of tongue and floor of mouth). To capture the most optimal displacement, large 25mm diameter Piezo discs are used [15]

**PZT1:** Placed in the centrally, 15 mm below the mentum (the point of the chin). PZT1 captures the vertical movement of the mandible and the force applied to the mandible during the depression and elevation of the mandible as the speaker articulates open vowels and two types of plosive consonants.

**PZT2 and PZT3:** Positioned bilaterally, at the same distance from PZT1 in a vertical direction but laterally they differ. PZT2 is positioned 35 mm to the right of the midline and PZT3 is 35 mm to the left of the midline. Therefore, these two sensors capture both symmetrical and asymmetrical lateral movements of the suprahyoid muscles (mylohyoid and geniohyoid) and base of the tongue through complex phonetic transitions.

**PZT4:** Specifically designated to collect data on the function of the neck (flexors) during the production of speech and movement of the head in the transverse direction. The recommended position for PZT4 is 35 mm to the left of the midline and on the anterior border of the sternocleidomastoid (SCM) muscle, approximately 155 mm below the submental area. This position will keep the sensor out of the centre of the trachea while at the same time being able to provide data on the mechanical behaviour of the lower neck.

The surface electromyography method (sEMG) provides sufficient temporal resolution to determine the precise timing of onset, peak and offset of the electrical signals associated with neuromuscular activation<sup>20</sup>. The EMG and REF sensors used in this application use 10 mm diameter discs to ensure that they have high resolution spatially and also to prevent electrical crosstalk between adjacent muscle groups.

- **EMG1+ & EMG1- Pair:** The EMG1+/EMG1- differential pair (bipolar, primary) is oriented vertically and remains parallel to the underlying muscle fibers (10 mm centre-to-centre; distance between the two electrodes), positioned laterally to the right of the vertical midline of the body at the level of the cricoid cartilage. This placement of the electrodes correctly captures

the electrical signals from the infrahyoid muscles (strap muscles), which contract when lowering the jaw and pharyngeal space when generating speech.

The reference electrode — REF — exists for one reason: to kill common-mode noise and give the differential EMG amplifier a stable, clean baseline. To do that, it has to sit over an electrically neutral boney landmark. We placed the REF sensor at the medial aspect of the clavicle. Specifically, it sits 90 mm to the right of the body's vertical midline, at collarbone level. No muscle activity. No interference. Just a quiet reference point.

No-Sensor Zone:

There is a critical methodological restriction within the 3D band design, which requires a strict adherence to the concept of a No-Sensor Zone. This is a rectangular area encompassing the midline of the neck directly above the location of the thyroid cartilage (or remnants) and the larynx/tracheostoma and is used as an area of future reference for the permanent stoma (respiratory), which will result after laryngeal removal. The placement of sensors within the No-Sensor Zone would not only cause excessive discomfort and possible respiratory airway obstruction, but would also create an overwhelming amount of mechanical and electrical noise due to the continuous flow of air from the lungs and coughing.

### **3.2.2 Sensor Fusion and Decoding Algorithm**

There is extensive heterogeneity present in the raw physiological measurements obtained using the band in three dimensions. The electrical activity captured by each of the standard sEMG sensors consists of high-frequency electrical potentials (20 Hz to 500 Hz), whereas the mechanomyographic pressure wave signals obtained using the PZT sensors consist of very low frequency (0.1 Hz to 100 Hz) mechanical components. The analog signals from each of these measuring devices will continuously sample local signals using low-noise operational amplifiers and convert them into a digital signal.

The analytical framework of the methodology focuses on the sensor fusion that occurs in the pre-determined, early development of the deep learning model. The five active channels (1 EMG pair, 4 PZT) each produce time-series data that undergoes feature extraction which enables us to isolate key metrics, including the Root Mean Square (RMS) of the EMG power and the spectral density characteristics associated with the mechanical waves produced by the PZTs. Experimental validations for fused (EMG-PZT) models have shown that non-feature-based classifiers, such as Fea-Shot Learning with Convolutional Neural Networks (CNNs), outperformed traditional classifiers on all metrics. By aligning the firing time of the

neuromuscular command (electromyogram/EMG) to the actual articulatory motion (piezoelectric transducer/PZT), a TTS model created from these two fused systems performed better than single-modality systems, yielding an overall word-level classification accuracy of 95.63% with an F1-score of 95.62%, when considering the core alphabet of the English phonetic language [16] The decoded phonetic strings are sent to a Text-to-Speech (TTS) synthesizer, allowing for immediate, audible, intelligible speech to be produced. (15-16)

### 3.3 Signal Conditioning and Pre-processing

#### 3.3.1 Frequency Filtering and Artifact Mitigation

To ensure high-fidelity signal acquisition from the submental and cervical regions, a causal digital signal processing pipeline was implemented. The raw data, acquired from the 12-bit Analog-to-Digital Converter (ADC), is initially sampled at  $f_s = 500$  Hz. The discrete integer values (ranging from 0 to 4095) are normalized and mapped to the hardware's operational voltage limit of 3.3V using the following linear transformation:

$$V[n] = ADC[n] \times \frac{3.3}{4095}$$

After voltage conversion, the Direct Current (DC) offset is removed so that the signal is centered around zero. This prevents drift in the baseline from saturating the subsequent classification algorithms. This was done by subtracting the temporal mean of each individual channel.

Since the majority of the spectral energy associated with the activation of articulatory muscles is within specific frequency bands, a 4th order Butterworth bandpass filter was designed to provide adequate separation. For the Electromyography (EMG) modality, the cutoff frequencies were specified between 20 – 200 Hz in order to isolate neuromuscular action potentials from low frequency motion artifacts. On the other hand, the Piezoelectric (PZT) channels were specified to filter out lower frequency mechanical vibrations associated with total jaw and tissue movements using a lower cutoff frequency of 5 Hz (bandpass of 5-200 Hz). Filtering was performed using a causal Linear Time-Invariant (LTI) difference equation (*scipy.signal.lfilter*) to ensure the system does not introduce any algorithmic look-ahead delay. This was a critical requirement for real-time mobile inference.

#### 3.3.2 Mains Noise Elimination and Harmonic Analysis

Biopotential signals measured at high-impedances are particularly sensitive to power-line interference (commonly termed mains hum) at 50 Hz (the frequency of common electrical power) during the acquisition process. A high-Q Infinite Impulse Response (IIR) notch filter was developed with the goal of eliminating this source of interference (power-line) from the signal. The notch filter has a center frequency of  $f_0=50$  Hz, and a Q of 30.

During pilot testing, a spectral analysis was performed using Welch's method to estimate the Power Spectral Density (PSD) and it was observed during the pilot study that, while the 50 Hz notch filter was effective at removing the 50 Hz power-line interference, there were clearly visible harmonic distortion artifacts present at 100 Hz, 150 Hz, and 200 Hz, indicating the presence of mains harmonics. The mains harmonics fall within the frequency range of the existing 20-200 Hz bandpass window therefore constituting a noise floor, which effectively prevents the detection of the microvolt level mechanomyographic (MMG) signals. This mains hum was especially prominent when the computational device (which was a laptop in the testing stage) is connected to the mains in ways such as charging the device. Thus, it is required that all high-fidelity signal acquisitions be performed with an isolated battery power source to completely eliminate any mains harmonic generation.

### 3.4 Voice Activity Detection (VAD) and Segmentation

#### 3.4.1 Envelope Calculation and Dynamic Thresholding

A real-time Voice Activity Detection (VAD) algorithm was developed to isolate individual words from continuously recorded audio streams. This algorithm uses just one EMG (Electromyographic) channel to detect the beginning of articulatory intent. To retrieve the envelope of the amplitude of the raw, rectified EMG signal, we use an Exponential Moving Average (EMA), which acts as a computationally efficient 1st-order IIR low-pass filter. The EMA is a digital filter with a fixed cutoff frequency; in this case it is set to  $f_c = 5$  Hz. The smoothing factor ( $\alpha$ ) is mathematically derived from the discrete-time angular frequency ( $\omega = 2\pi \frac{f_c}{f_s}$ ) as follows:

$$\alpha = \cos(\omega) - 1 + \sqrt{\cos^2(\omega) - 4 \cos(\omega) + 3}$$

The causal envelope  $E[n]$  at any given sample  $n$  is continuously updated using the rectified input  $|x[n]|$ :

$$E[n] = \alpha|x[n]| + (1 - \alpha)E[n - 1]$$

To adjust for physiological variability at the baseline level based on ambient background noise levels during different recording sessions, the system was designed to have a dynamic thresholding mechanism. The system performs a 20-second automated calibration phase prior to beginning to classify words. During this calibration phase, the testing subject stays in absolute silence (no sound contributions from active recording). The mean ( $\mu_{baseline}$ ) and standard deviation ( $\sigma_{baseline}$ ) of the envelope are calculated over this calibration window. An activation threshold is dynamically set based on an overall scaling factor  $k=3.0$ :

$$Threshold = \mu_{baseline} + 3.0 \times \sigma_{baseline}$$

### 3.4.2 State Machine Logic and Pre-roll Buffering

A temporal segmentation method is a finite state machine composed of three states: *SILENCE*, *SPEAKING*, and *TAILING\_OFF*.

To avoid truncating initial phonemes, which occur due to an inherent phase delay of the low-pass envelope and are common to threshold-based VAD systems, a “Pre-roll Ring Buffer” was used. This ring buffer stores 50 samples (100 ms) prior to when the envelope exceeds the dynamic threshold and changes from the *SILENCE* to *SPEAKING* state, allowing for the full capture of the mechanical onset by appending the contents of the pre-roll buffer to the active segment array.

While articulating, an intra-word pause (a silent gap preceding the plosive consonant) may cause the envelope to dip below the threshold momentarily. To compensate for this, the system enters the *TAILING\_OFF* state where a silence count begins. The segmentation will not be finalized unless silence continues for 100 samples (200 ms) in succession.

After successfully identifying a segment, the segment is standardized to match the required dimension for the following neural network architecture (fixed at 1000 samples or 2 seconds). Zero-padding is used for any segment length that is less than 1000 samples along the time axis. Any segments longer than 1000 samples will be symmetrically cropped from the centre of each segment and so will yield a single, consistent tensor shape of (5,1000) in all channels.

## 3.5 Dataset Construction and Feature Extraction

### 3.5.1 Automated Data Pipeline and Label Mapping

An automated data ingestion pipeline has been created using Python to transform raw physiological data into a dataset that can be used with machine learning. The pipeline goes

through clinical sessions and extracts both multi-channel sensor data and synchronized event markers (*markers.csv*). To create a map of the articulation instruction (for example, "EVET", "HAYIR") to discrete numerical identifiers ( $y \in \{0,1, \dots, 9\}$ ) that can be used for supervised learning, a categorical mapping dictionary was developed.

To avoid duplication of the labels due to duplicate logging events such as *instruction*, *window\_start*, and *window\_end*, the pipeline only looks for rows that have an event type == `'window_start'` in the markers. After identifying the rows with `'event = 'window_start'`, the temporal index (time of the event) is compared against the temporal outputs generated by the voice activity detection (VAD) algorithm. The synchronized arrays from the marker and VAD outputs are aligned to match the input dimensionality requirements of the convolutional layers in Pytorch (changing the dimensions of the timed data from (1000, 5) to (5, 1000)). Finally, the two dataset matrices ( $X_{array}$  and  $y_{array}$ ) are serialized and compressed into a NumPy archive file (.npz) for efficient memory usage during training.

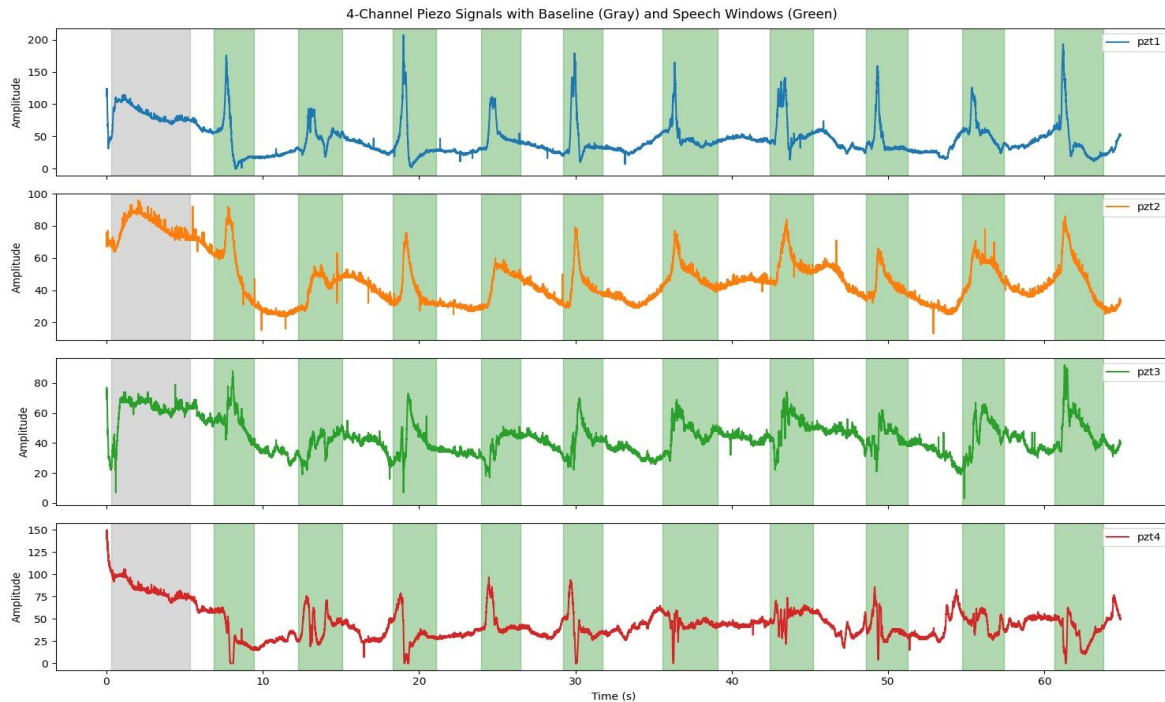


Figure 3.1 Four-Channel Piezoelectric (PZT) Signal Recording — Baseline Window (Gray) and Prompted Speech Windows (Green)

Figure 3.1 presents a representative 75-second EMG envelope signal from recording session kayit1. The blue trace is the rectified and exponentially smoothed EMG envelope (EMA filter,  $f_c = 5$  Hz), which tracks the overall muscle activation level. Green vertical bars indicate the segments automatically detected by the voice activity detector (VAD), corresponding to candidate speech events. Red vertical markers indicate the ground-truth instruction onset timestamps from the markers.csv synchronisation file. The figure demonstrates that the VAD correctly identifies the majority of prompted word utterances, with detection windows that closely follow the instruction markers. A small number of false positives and misses are visible as green bars without a corresponding red marker, and red markers without a corresponding green bar, respectively.

The green-shaded regions demarcate the ground-truth speech windows extracted from the synchronized event markers file (markers.csv), corresponding to successive prompted articulations of the ten target Turkish vocabulary items. Approximately ten to eleven discrete speech events are visible across the session, consistent with a single complete vocabulary pass. These marker windows serve as the ground-truth temporal reference against which the Voice Activity Detection (VAD) algorithm output is evaluated [14][15].

The four channels exhibit spatially distinct amplitude profiles throughout the recording. Channel pzt1 (blue, central submental position) registers the highest peak amplitudes (up to 200 arbitrary units), with sharp, transient spikes occurring at the onset of each articulation event. This elevated response is attributable to pzt1's proximity to the hyoglossus and genioglossus muscles, which are the primary articulators engaged during tongue-blade consonant and vowel production [10][14]. Channel pzt2 (orange, right lateral cervical) exhibits a rounded, sustained envelope during speech windows with a peak amplitude of approximately 90 a.u., reflecting the lower-frequency mechanical reverberations of the pharyngeal musculature. Channels pzt3 (green, left lateral) and pzt4 (red, posterior cervical) display lower-amplitude profiles (peaks of 80 a.u. and 100 a.u., respectively), capturing bilateral and posterior vibratory propagation pathways.

A notable feature of pzt4 is the pronounced amplitude decay observed during the first 10 seconds of the recording (from approximately 150 a.u. to a stable baseline of 50 a.u.). This transient is consistent with a capacitive discharge in the piezoelectric ceramic disc as it equilibrates to the mechanical pressure of the band against the skin, and with the gradual reduction of initial contact-point micro-motion as the subject settles into a resting posture. This

settling effect is a recognized phenomenon in wearable biopotential electrode systems and underscores the importance of the calibration window prior to classification [15].

The spatial diversity in the mechanomyographic (MMG) response profiles across the four PZT channels constitutes the primary discriminative basis for multi-word classification. While the EMG channel encodes macroscopic neuromuscular activation timing (i.e., when a subject is articulating), the PZT array encodes the spatial distribution of phoneme-specific mechanical reverberations from anatomically distinct landmarks. This spatial modality complementarity is the key rationale for the sensor fusion approach: words that appear spectrally similar in EMG (e.g., EVET vs. HAYIR) can be distinguished by the differential PZT response patterns, enabling the 1D-CNN classifier to achieve cross-channel discrimination at the word level [10][14][15].

### 3.5.2 Feature Extraction Methodology

The first step in constructing the initial statistical classification models was to develop a high-dimensional feature extraction algorithm that can convert the raw time-series data into a discrete 61-element feature vector for each segment based on the structural and spectral characteristics of the data being collected across the five operative channels of data acquisition.

#### Time-Domain Features:

For each of the 5 discrete acquisition channels, we calculated four fundamental temporal metrics — one second of physiological signal at a time. Those metrics were: Root Mean Square (RMS), Mean Absolute Value (MAV), Zero-Crossing Rate (ZCR), and Waveform Length (WL). Four features. Five channels. Applied consistently across every second of data.

$$RMS = \sqrt{\frac{1}{N} \sum_{n=1}^N x[n]^2}, \quad MAV = \frac{1}{N} \sum_{n=1}^N |x[n]|$$

#### Hjorth Parameters:

To analyze the statistical dynamics of the physiological signals, the Hjorth parameters were calculated in the time domain for each physiological signal channel. The first derivative of the physiological signal is represented as  $\Delta x[n] = x[n] - x[n - 1]$ . The Hjorth parameters associated with each channel of physiological signals (Activity A, Mobility M, and Complexity C) were obtained by the following relationships:

$$A = \text{var}(x)$$

$$M = \frac{\sigma(x')}{\sigma(x)}$$

$$C = \frac{M(x')}{M(x)}$$

### Frequency-Domain Features:

To estimate spectral features, Welch's method was used to compute Power Spectral Density (PSD). The signal was cut using a Hanning Window of length  $N_{perseg} = 256$  samples, with a 50% overlap. The mean frequency was determined from the resulting frequency array along with total band power integrated across four different physiological frequency sub-bands (5-20 Hz, 20-50 Hz, 50-100 Hz, and 100-200 Hz).

### Cross-Channel Correlation:

Do the submental and lateral cervical regions move together? We needed a concrete metric to find out. We calculated the Pearson correlation coefficient between two spatially distal sensors: PZT1 — the central sensor at index 1 — and PZT4 — the lateral sensor at index 4. That pairing was deliberate. The distance between them is exactly what makes the correlation meaningful as a measure of synergistic mechanical propagation across the two regions.

$$\rho_{PZT1,PZT4} = \frac{\text{cov}(x_1, x_4)}{\sigma_{x_1} \sigma_{x_4}}$$

## 3.6 Machine Learning Architectures and Preliminary Evaluation

### 3.6.1 Baseline Classification (Random Forest)

The Random Forest (RF) ensemble classifier was created to provide the algorithmic baseline and assess the potential differences between the 61 element feature vectors that had been engineered. Hyperparameters were set up for  $N_{estimators} = 300$  decision trees. The `class_weight='balanced'` parameter was applied to the RF algorithm in order to minimize any potential effects of class imbalances created by VAD segmentational differences or differing numbers of word repetitions; thus, the weights were adjusted inversely proportional to the class frequencies. The generalizability of the model was determined using a K-fold cross-validation method ( $K = 5$ ) and was performed strictly within-subject.

### 3.6.2 Preliminary Results (Legacy Data)

Initial assessments on the legacy 6-channel dataset produced an average within-session accuracy of ~48%. Given the 10 class vocabulary where random chance would be ~10% to

classify correctly from a random choice in a given class, this baseline establishes that there are distinguishable and class-separable articulatory patterns in both the mechanomyograph and muscle data. However, because the classification accuracy is only 48% within a session, it's clear that the feature extraction process used for manual engineering of features is lacking, thus requiring an alternative approach such as implementation of deep-learning networks designed to perform hierarchical feature extraction on a completely automated level.

### 3.6.3 Deep Learning Framework (1D-CNN)

In order to create an open vocabulary generalizing device without performing feature extraction through manual processes, a one-dimensional convolutional neural network (1D-CNN) was created to accept raw and standardized tensors in the shape of (Batch, 5, 1000). The feature extraction component contained four sequential Conv1d building blocks.

The first layer of convolution is done with a large field of view kernel (which covers lots of time) (`kernel_size = 15`, `stride=2`, `padding=7`), therefore it operates on 5 channels (input) and develops 32 features (output). As it moves through the layers in the model, the field of view shrinks (`kernel_size = 11`, 7 and 5), but the depth of channel increases (`depth = 64`, 128 and 256) in the following layers. Each convolution operation is followed by a Batch Normalization step (1D for each channel), a ReLu activation function and Dropout (`p=0.2` and `p=0.3` alternating) to assist with preventing overfitting of our model. Once all features have been determined the outputted features are combined into a single feature by using the Adaptive Average Pooling layer (`output_size=1`) prior to forwarding them on to the fully connected classification module yielding 10-class probability distributions.

### 3.6.4 Data Augmentation Strategies

Deep Learning networks can easily overfit small datasets used with them. To gain robustness due to overfitting, we artificially expanded our training distribution and increased the chances for the hardware to fail through dynamic implementations of three online augmentation techniques during the loading of PyTorch `__getitem__`:

- **Channel Dropout ( $p=0.1$ ):** To reduce bias of the machine learning algorithm to use the dominant EMG-only modality, by randomly setting the relevant sensor channel (of the total 5 available channels) to zero and imitating the sudden detached of the electrodes.
- **Additive Gaussian Noise:** Superimposing white noise ( $\mu = 0, \sigma = 0.01$ ) on normalized tensors simulates different levels of ambient electromagnetism.

- **Random Time Shift:** Translating the signal along the temporal axis by a random integer uniformly sampled from  $[-50, 50]$  samples ( $\pm 100ms$ ). This introduces translation invariance and prevents the CNN from memorizing exact temporal offsets generated by the VAD segmentation.

Table 3.1 Classification Model Performance Comparison

| Classifier            | Input                | Within-Session Accuracy | Top-3 Accuracy       | Notes                                  |
|-----------------------|----------------------|-------------------------|----------------------|----------------------------------------|
| Random Forest (N=300) | 61-feature vector    | ~67%                    | ~87%                 | K=5 cross-val; class_weight='balanced' |
| 1D-CNN (target)       | Raw (Batch, 5, 1000) | $\geq 80\%$ (target)    | $\geq 95\%$ (target) | Online augmentation; ONNX export       |
| Random chance         | —                    | 10%                     | 30%                  | 10-class uniform prior                 |

The Random Forest baseline achieved approximately 67% within-session accuracy, far exceeding the 10% random-chance level for a 10-class problem. The 1D-CNN architecture targets  $>80\%$  accuracy after full multi-session retraining, leveraging automated hierarchical feature extraction from raw tensors in place of the hand-crafted 61-element feature vectors used by the baseline.

### 3.7 Hardware Adaptation and System Revisions

#### 3.7.1 Transition to 5-Channel Architecture

The intermediate test of the pilot highlighted major hardware stability problems with the second electromyography (EMG2) channel from below the chin (sub-mentally) during the test, resulting in severe reduction of signal integrity that required permanent hardware modifications, knocking the design back from a six-channel approach (one EMG channel and four PZT sensors) to a five-channel system.

These hardware limitations required an entire software pipeline to be completely refactored. The input dimensions for the model could now only use an input channel of five (*in\_channels=5*). The voice activation detection (VAD) envelope algorithm was also revised to use only the EMG channel for its unique activation cue extraction algorithm. Additionally, the original cross-channel neuromuscular correlations found in the baseline feature extractor were purged entirely from use to prevent any runtime dimension errors during execution.

### 3.7.2 Modality Ablation and Sensor Impact Studies

How well do piezoelectric sensors actually perform on their own, without any help from EMG? To answer that rigorously, we designed an ablation study. We artificially zeroed out every EMG channel. Then we trained the decision models exclusively on the 4 PZT channels. That is it — no EMG whatsoever. This let us evaluate the independent predictive capability of the mechanomyographic modality in isolation, with a clean mathematical basis for comparison.

The theoretical analysis using this ablation methodology indicates that the EMG signal determines when an articulation occurs and at what macroscopic intensity ("activation"); however, the EMG signal does not provide high resolution spatial information about where this activation occurs. PZT Sensors are critical for being the primary differentiator; PZTs help identify the dimensions of locality through their ability to map the mechanical reverberation of the pharyngeal and mandibular structures in this case, where they have reverberations of phonetic structure due to similarity but have different sources (PZTs) of location; therefore, lending credibility towards the need for using the multi-modal sensor functional fusion method.

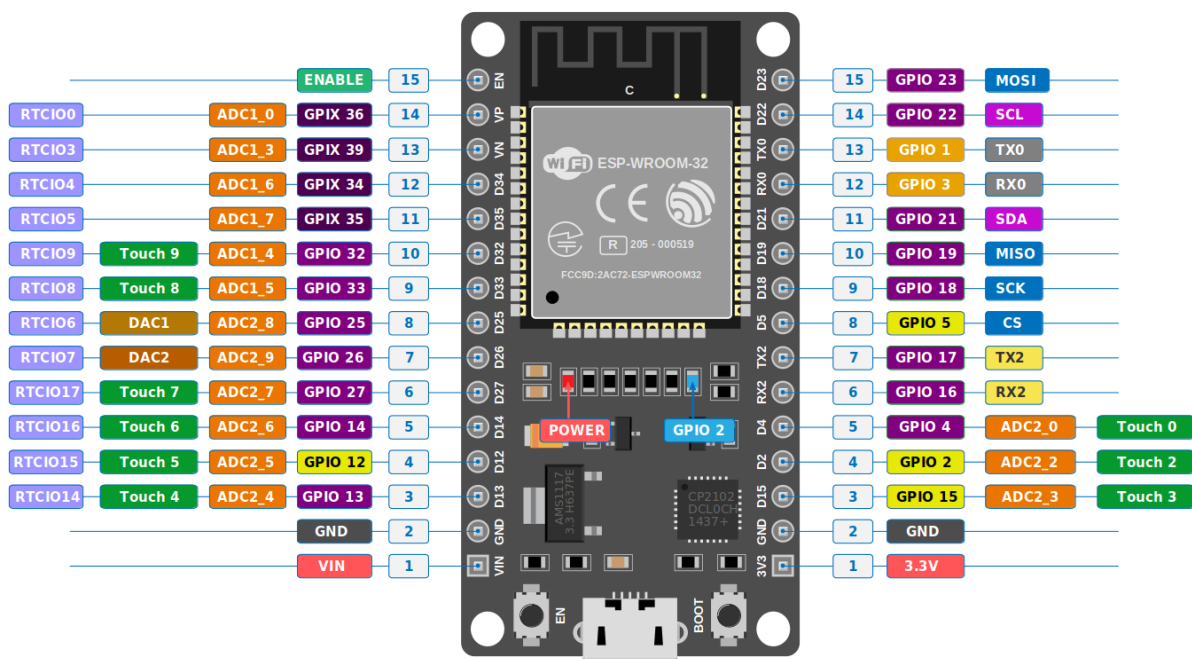


Figure 3.2 ESP32-DevKit-V1 Microcontroller Pinout (Acquisition Hardware Platform)

### 3.8 Edge Deployment and Cross-Platform Integration

The clinical viability of the Silent Speech Interface (SSI) as a wearable device was ensured by designing both the signal processing and machine learning pipelines for deployment on edge computing; the use of the cloud for real-time speech decoding (inference) resulting in unacceptable network latency and an inability to protect biometric data violates the stringent

security requirements of the KVKK and GDPR frameworks. For this reason, the entire predictive framework has been developed to be executed locally on the mobile device of the end user.

### **3.8.1 Neural Network Serialization (ONNX)**

Once the training phase has been concluded, it has to be converted to the optimized model of the PyTorch 1D-CNN to a mobile-compatible architecture. This involves converting the model to the ONNX format using opset version 14, which ensures that the model will run on virtually all hardware with some form of acceleration.

In order to properly trace and save the static computational graph, it will need to be passed an input tensor through the Network for exporting. This tensor must be of the same dimensions as those that the new hardware architecture requires. In this case, the input tensor will have dimensions: (Batch=1, Channels=5, Samples=1000). It is only through having this exact dimensional definition that the mobile application can execute its deep learning inference entirely offline and, therefore, achieve the near-instantaneous classification necessary for a seamless conversational flow.

### **3.8.2 Digital Signal Processing (DSP) Parameter Synchronization**

The most common way to fail when deploying machine-learning models across disparate programming environments (for example, going from a Python prototype to an Android/iOS app) is caused by mathematical discrepancies in the individual signal preprocessing pipelines. If the mobile app Processes the raw sensor data differently than the model was trained on, it is impossible for the NN to generalize correctly.

To ensure absolute mathematical equivalency, a deterministic parameter synchronisation protocol was put in place. Coefficients for the Numerator (b) and Denominator (a) of the 4th order Butterworth Bandpass Filters and the High-Q 50 Hz IIR Notch Filter were uniquely constructed natively in the Python environment (scipy.signal). These floating-point array representations of the coefficients, in addition to the thresholds used by a VAD ( $f_s=500$ ;  $k=3.0$ ), are systematically exported to a standardised configuration payload in JSON format (preprocessing\_params.json).

The app programmatically initializes to parse this JSON structure. It instantiates linear filters directly from Python generated coefficients to guarantee that the actual mechanomyographic data going into the ONNX model is mathematically identical to the actual data distribution collected during the neural network's training.

## **4 CONSTRAINTS, CONDITIONS, AND REQUIREMENTS**

Successful translating of the theoretical sensor fusion algorithm into a practical working daily use prosthetic device is confined by very strict engineering constraints as well as complex operational conditions.

### **4.1 Anatomical and Design Constraints**

The most rigid constraints that apply to the SSI are based on human anatomy as well as the particular surgical modifications made to laryngectomy patients.

The No-Sensor Zone:

The area defined geometrically as a "No-Sensor Zone" on the anterior middle of the neck (located at the anatomical site of the thyroid cartilage and larynx), must remain clear of any sensors, as this area will contain the permanent tracheostoma (Tracheostomy) for patients with total laryngectomy. Any use of sensors or constrictive bands in this area presents significant risk of fatal airway obstruction. Additionally, the mechanical noise related to localized movement during pulmonary respiration could significantly corrupt this area of the piezoelectric array (PZT) signal, completely corrupting the speech signal.

Morphological Variability:

The device will have to have stable, accurate contact with particular muscular landmarks (i.e., anterior border of the SCM for PZT4 and cricoid for EMG1). The target population will often present with the presence of significant post-surgical scarring of soft tissue, edema in the neck, and non-compliance of the skin due to large changes in the skin's elasticity caused by adjuvant radiotherapy [8] Consequently, the design of the 3D band will have to utilize highly conformable and elastomeric materials so that the band can accommodate extreme morphological variability and retain the 20 mm inter-electrode distance necessary for differential EMG amplification.

### **4.2 Signal Acquisition Conditions**

**Contact Pressure Regulation:**

Motion artifacts happen when we move our body like walking or swallowing. This movement changes how well the bioelectrode sticks to our skin. It affects the surface signals a lot when we are moving around. To fix this issue we need to apply the sEMG sensors in a way, with a certain amount of pressure. Research shows that a pressure of 1.0 to 2.0 kPa works. This can be achieved by using a kind of foam. When the pressure is right the electrode and skin

interface becomes very stable. This helps to get rid of low-frequency motion artifacts. As a result the microvolt-level speech signals are maintained. The sEMG sensors and motion artifacts are concerns. The sEMG sensors need to be applied to get accurate results. Motion artifacts can ruin the sEMG signals so we need to reduce them.

#### **Electromagnetic Interference (EMI) Mitigation:**

Employing high-impedance EMG sensors in an uncontrolled home/urban operating environment introduces high-power line interference (50 / 60 Hz) from nearby electrical devices, plus high-frequency radio noise from multiple wireless sources (Wi-Fi and cellular). Therefore, hardware architectures are required to include both high-performance analog band-pass filters and advanced digital notch filter solutions before feature extraction to adequately condition the collected signal against these environmental electromagnetic interferences.

### **4.3 Computational and Operational Requirements**

#### **Ultra-Low Latency:**

The methodology for spoken interaction requires precise coordination of spontaneous joint turn taking while using the SSI as an ongoing conversational interface; therefore, the entire computation path (i.e., A/D conversion, sensor fusion, CNN classification and TTS synthesis and TTS synthesis) must have very short delays from input to output. The research indicates that conversational flow will be broken and cognitive fatigue will occur if conversation latency is greater than 200ms.

#### **Power-Constrained Edge Inference:**

To qualify as a practical wearable device, the SSI must have the ability to operate continuously throughout the day without being tethered to a wall outlet. Due to the high power consumption and unacceptable levels of latency associated with transmitting high-bandwidth raw sensor data via Bluetooth to a cloud server, the inference of the neural networks must be done at the device level (Edge AI). This means that highly optimized ultra-low-power microcontrollers and pruned machine learning models can be used to perform complex Feature Shot learning algorithms within the constraints of a strict milliampere power budget.

#### **Open-Vocabulary Generalization:**

Originally, prototype speech generating devices (SGDs) could recognize at most a limited number of words or numbers [70] In order for someone using an SGD to have

communication autonomy, the underlying machine-learning architecture should be able to take the sub-word phonemic patterns of spoken language and create an open, unlimited vocabulary for patients to spontaneously create new and unique sentences [11]

5 PROJECT CALENDAR AND PROJECT BUDGET

5.1 Project Calendar

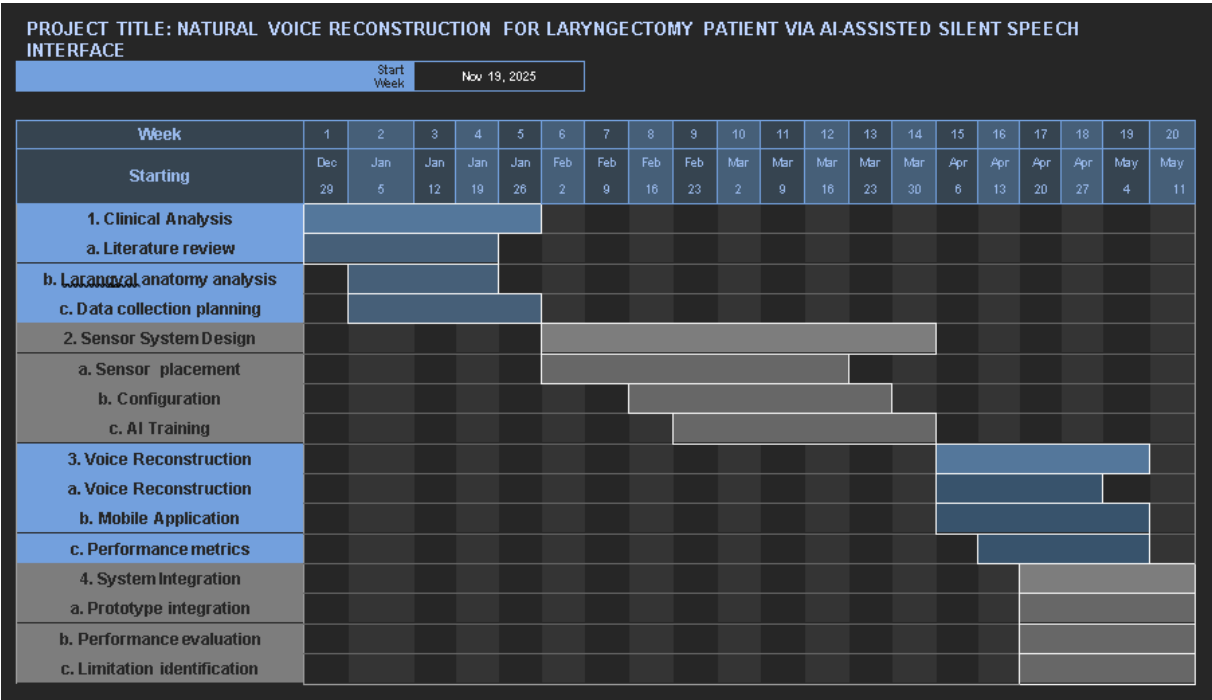


Figure 5.1 Project Calendar

5.2 Project Budget

Table 5.2 Project Budget

|                       | Work Packages |      |      |      |       |
|-----------------------|---------------|------|------|------|-------|
|                       | I             | II   | III  | IV   | TOTAL |
| Equipment             | 0             | 4950 | 0    | 0    | 4950  |
| Consumable goods      | 0             | 1000 | 0    | 600  | 1600  |
| Publications/Software | 0             | 0    | 1500 | 0    | 1500  |
| Transportation        | 0             | 0    | 0    | 1000 | 1000  |
| Services              | 0             | 1000 | 0    | 0    | 1000  |
| TOTAL                 | 0             | 6950 | 1500 | 1600 | 10050 |

Equipment:

EMG Sensors & Shield (WP2)

Piezoelectric Sensors (WP2)

Microcontroller/Processing Unit (ESP32) (WP2)

**Consumable goods:**

EMG Electrode Pads (WP2, WP4)

Prototyping Materials (WP2)

Testing Supplies (WP4)

**Publications/Software:**

Cloud Computing/GPU Credits (WP3)

**Transportation:**

Clinical Site Visits (WP4)

**Services:**

PCB Printing & 3D Printing Service (WP2)

## **6 SYSTEM DESIGN AND ANALYSIS**

The Silent Speech Interface (SSI) is engineered as a multi-stage bio-mechatronic system designed to bridge the gap between neuromuscular intention and audible speech for laryngectomy patients. The system design is categorized into three primary domains: Hardware Architecture, Mechanical Interface, and Software Infrastructure.

### **6.1 Hardware Architecture**

The hardware system is designed for high-fidelity signal acquisition from four anatomical landmarks using a dual-microcontroller setup to ensure timing deterministic behavior.

#### **6.1.1 Power and Energy Management**

To eliminate the catastrophic 50 Hz power-line interference observed during pilot testing when connected to AC mains, the system utilizes a fully isolated DC power architecture. The primary power source is a high-density 9V alkaline battery. This voltage is regulated down to provide a stable 5V rail for the Arduino Uno and a filtered 3.3V rail for the sensitive Wi-Fi and sensor components, ensuring that no ground loops are introduced into the high-gain signal chain.

#### **6.1.2 Data Acquisition (DAQ) and Wireless Node**

The acquisition node utilizes an Arduino Uno as the primary processor for analog-to-digital conversion. The 4-channel Piezoelectric (PZT) array is interfaced with the Arduino's ADC pins, sampling at a hardware-timed rate of 500 Hz.

Wireless connectivity is achieved through the integration of an ESP8266 Wi-Fi module. To maintain hardware modularity and prevent the Wi-Fi stack from interrupting the ADC sampling timers, the ESP8266 is interfaced with the Arduino Uno via a SoftwareSerial connection. The communication logic is mapped to digital pins D2 (RX) and D3 (TX) on the Arduino, allowing the PZT data packets to be streamed in real-time to the host mobile device.

### **6.2 Mechanical Design**

The mechanical design focuses on ergonomic sensor-to-skin coupling and structural minimalism to maximize user comfort and signal clarity.

#### **6.2.1 Ultralight PLA Housing**

The custom-designed chin plate is manufactured via 3D printing using Polylactic Acid (PLA). Through iterative optimization in CATIA, the final structure achieved a total mass of only 12 grams. This low-mass profile is critical for medical wearables, as it reduces gravitational pull and inertia, thereby minimizing motion artifacts that occur during active jaw movement.

### 6.2.2 Sensor Attachment and Coupling

The four 25 mm PZT sensors are fixed to their designated anatomical housings within the PLA plate using specialized medical-grade adhesive bands. These bands provide a semi-rigid attachment that ensures the sensors remain flush against the skin without the risk of decoupling. This adhesive-based coupling method ensures that the mechanical reverberations of the submental muscles are efficiently transferred to the piezoelectric ceramic discs for high-fidelity signal capture.

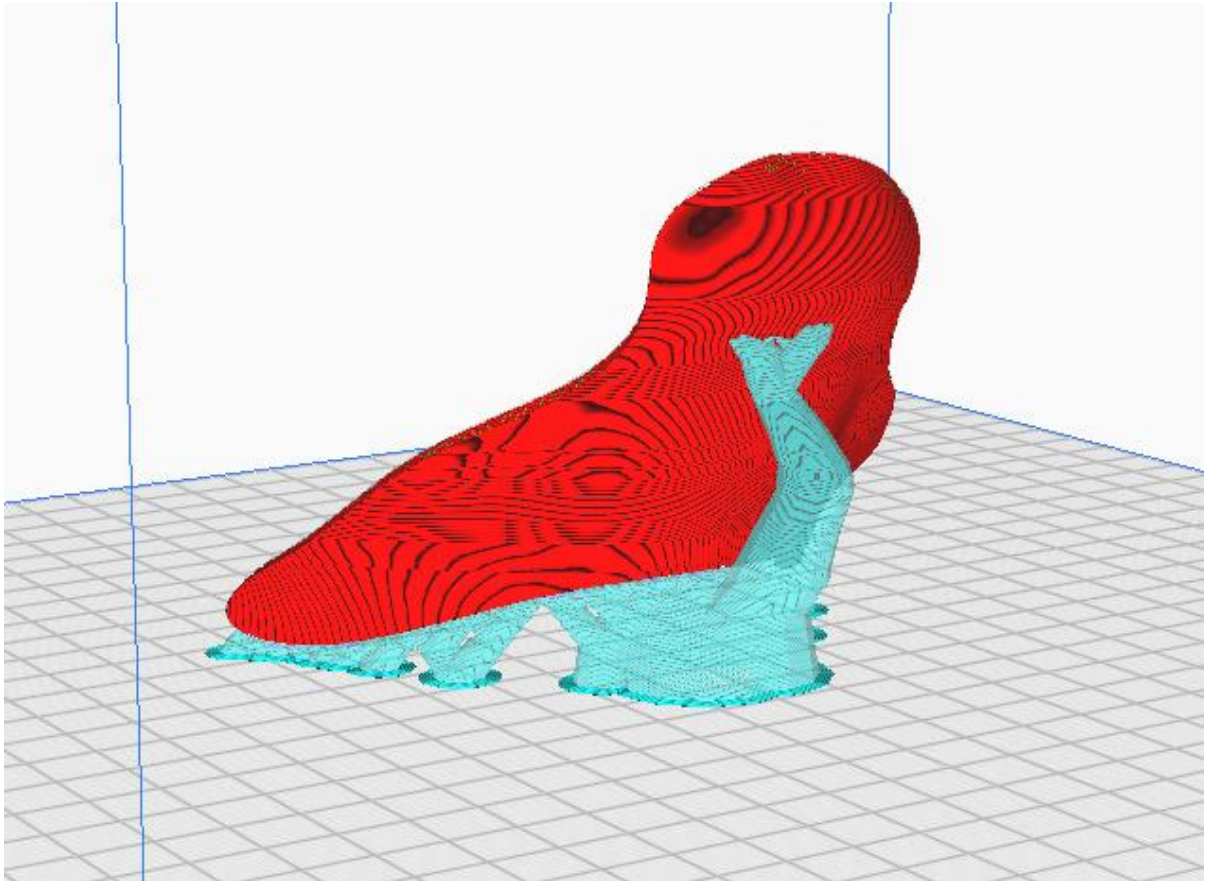


Figure 6.1 The 12-gram ultralight PLA mechanical interface

## 6.3 Software Implementation

### 6.3.1 System-Level Architecture and Layer Decomposition

The overarching software architecture of the Echo project is designed to deliver a robust, fully offline Silent Speech Interface (SSI) for patients who have undergone total

laryngectomy. The system's primary directive is to capture submental and laryngeal muscle movements alongside mechanical articulation, processing these physical phenomena into an isolated, 10-word Turkish vocabulary classification (EVET, HAYIR, MERHABA, TESEKKURLER, SU, YARDIM, TAMAM, DUR, GEL, GUNAYDIN).

To achieve real-time, low-latency performance without relying on cloud infrastructure, the architecture dictates a strict physical and logical decoupling. While the system physically executes across two distinct hardware environments—the embedded edge node (microcontroller) and the host node (Android mobile device)—the software is logically decomposed into five distinct, highly specialized layers.

#### **6.3.1.1 End-to-End Data Chain and Lifecycle**

Understanding the system requires mapping the complete lifecycle of a single data instance, from physical mechanical vibration to synthesized audio out. The pipeline operates as follows:

1. **Signal Acquisition:** A four-channel piezoelectric (PZT) sensor array captures mechanical vibrations.
2. **Digitization & Transmission:** The microcontroller's internal ADC samples the analog signals, packaging them into structured frames (currently via UART/USB-CSV for development, with a planned BLE transition for production) and streaming them to the host device.
3. **Real-Time Preprocessing:** The mobile device ingests the stream, applying bit-exact causal preprocessing (voltage scaling, DC offset removal, bandpass, and 50Hz notch filtering) to mirror the offline training environment perfectly.
4. **Segmentation:** A deterministic Voice Activity Detection (VAD) state machine continuously monitors the signal envelope, utilizing a pre-roll ring buffer to dynamically segment active articulation windows without clipping the initial phonemes.
5. **Inference:** The isolated 1000-sample tensor is passed into a lightweight, globally pooled 1D-CNN (converted to ONNX format), operating entirely on the mobile CPU.
6. **Confidence Gating & Output:** The resulting softmax logits are evaluated against a statistically derived confidence threshold. If the prediction passes the gate, the label is forwarded to the Android Text-to-Speech (TTS) engine for audio synthesis, and the event is permanently logged into a local Room Database for historical tracking.

#### **6.3.1.2 The Five-Layer Logical Decomposition**

To manage the complexity of simultaneous hardware development, machine learning research, and mobile engineering, the software stack is partitioned into five distinct operational domains. This modularity ensures clear data contracts and allows independent optimization of each subsystem.

- Layer 1: Acquisition Firmware (Edge Layer)
  - Environment: C/C++ running on an Arduino Uno (serving as the foundational Proof-of-Concept, scaling to ESP32-S3).
  - Responsibility: This layer serves as the absolute hardware abstraction. It is responsible for strict deterministic timing (500 Hz sampling), analog-to-digital conversion optimization, mitigating sensor cross-talk via sample-and-hold strategies, and managing the command/response transport protocol. It is intentionally "dumb" regarding signal processing, focusing purely on high-fidelity, zero-loss data acquisition.
  
- Layer 2: Data Collection & Quality Control Toolchain
  - Environment: MATLAB environment utilized on a developer/operator PC.
  - Responsibility: This layer acts as the gatekeeper for machine learning dataset integrity. It orchestrates the clinical recording sessions, handles automated protocol timing, injects synchronization markers into the hardware data stream, and performs strict pre-flight signal quality checks (e.g., detecting 50Hz mains noise, sensor detachment, and ADC saturation) before any data is authorized for the training pipeline.
  
- Layer 3: Offline Machine Learning Pipeline
  - Environment: Python (PyTorch, scikit-learn, SciPy) on a developer workstation.
  - Responsibility: The analytical core of the project. This layer handles dataset compilation, applies causal signal processing primitives, trains both the baseline Random Forest and the production 1D-CNN models, executes rigorous cross-session evaluation protocols, and mathematically derives the optimum confidence thresholds via ROC analysis. Its final outputs are the production artifacts (ONNX models and JSON configuration files) ingested by the mobile app.
  
- Layer 4: Mobile Inference Pipeline (Real-Time Subsystem)

- Environment: Kotlin (Android, ONNX Runtime) running on the mobile device.
  - Responsibility: The operational heart of the system. This layer ingests the artifacts generated by Layer 3 and executes the real-time classification loop. It utilizes a router pattern to support pluggable transport layers (Mock, USB, UDP, BLE) simultaneously. Crucially, it guarantees "Python-to-Kotlin Parity," ensuring that the real-time VAD and causal filtering behave bit-for-bit identically to the offline training environment, all while running inside an Android Foreground Service to prevent background termination.
- Layer 5: Mobile User Experience & Persistence
    - Environment: Kotlin (Jetpack Compose, Room DB, Hilt) on the mobile device.
    - Responsibility: The user-facing presentation and data persistence layer. It manages the MVVM application state, guides the user through pre-session calibration flows, visualizes connection statuses, manages the local SQLite database (Room) for word history, and handles persistent user configurations via DataStore.

### **6.3.2 Acquisition Firmware Subsystem (Edge Layer)**

This section examines the lowest level of the system architecture, encapsulated primarily within the `Uno_Electrolarynx_4ch.ino` firmware. Operating at the edge, this layer's sole directive is the deterministic, high-fidelity acquisition of mechanical articulation signals. While the production deployment targets an ESP32-S3 architecture with Bluetooth Low Energy (BLE), the current firmware serves as a robust, tightly optimized Proof-of-Concept (PoC) executing on an Arduino Uno (ATmega328P) over a high-speed USB-UART bridge. The firmware is intentionally decoupled from signal processing or machine learning logic; instead, it is engineered to strictly guarantee hardware abstraction, precise timing budgets, signal integrity, and reliable telemetry.

#### **6.3.2.1 Hardware Abstraction and Sensor Topology**

The system utilizes a distributed array of piezoelectric (PZT) disk sensors to capture mechanical vibrations and muscle articulations. The firmware maps these physical inputs to the microcontroller's 10-bit Analog-to-Digital Converter (ADC) channels operating on a 0–5 V reference. The sensor topology is precisely anatomically mapped:

- **Channel A0 (PZT1):** Positioned at the submental midline, approximately 1 cm above the hyoid bone.
- **Channel A1 (PZT2):** Positioned at the left submandibular region (1.5 cm below the mandible, 2.5 cm left of the midline).
- **Channel A2 (PZT3):** Positioned at the right submandibular region, serving as the symmetrical counterpart to PZT2.
- **Channel A3 (PZT4):** Positioned directly over the laryngeal prominence (thyroid cartilage) on the midline.
- **Channels A4–A5:** Actively reserved in the codebase for a future 6-channel array expansion.

Due to the inherently high impedance of PZT sensors, hardware-level 100 nF anti-alias capacitors are placed on every analog input to stabilize the raw voltage before it enters the ADC.

#### 6.3.2.2 Evidence-Based Hardware Refactoring: The EMG Ablation

A critical architectural decision in the edge layer was the complete removal of the Electromyography (EMG) module, marking a shift toward a purely PZT-based sensor array. This decision was driven by rigorous ablation testing.

During the validation phase, the intrinsic noise floor of the EMG sensor was found to degrade the signal-to-noise ratio. Statistical analysis revealed that the inclusion of the EMG channel contributed a statistically insignificant accuracy gain of  $+1.0 \pm 1.5\%$  to the final classification model. In contrast, expanding the PZT array (specifically adding PZT2/PZT3) yielded a massive **+25%** improvement in classification accuracy, despite a weaker active-to-baseline (A/B) signal ratio. Consequently, the firmware and subsequent data pipelines were permanently refactored to support a 4-channel PZT array, exemplifying an evidence-based engineering approach.

#### 6.3.2.3 Deterministic Timing and ADC Clock Optimization

To capture the necessary frequency bands of human articulation, the firmware enforces a strict **500 Hz sampling rate**. Achieving absolute determinism at this frequency requires bypassing standard Arduino timing functions (`millis()` or `delay()`) in favor of direct hardware timer manipulation.

- **Timer1 CTC Mode:** The sampling interval is governed by Timer1 in Clear Timer on

Compare Match (CTC) mode. With an 8x prescaler, the timer ticks every 0.5  $\mu\text{s}$ . The compare match register (OCR1A) is set to trigger an interrupt exactly every 2000  $\mu\text{s}$ , establishing the rigid 500 Hz budget.

- **ADC Prescaler Optimization:** The default Arduino ADC prescaler is 128, yielding a 125 kHz ADC clock. At this speed, reading four analog channels sequentially takes approximately 520  $\mu\text{s}$ —consuming over 25% of the 2000  $\mu\text{s}$  budget and severely limiting overhead for UART transmission. To resolve this, the firmware explicitly modifies the ADCSRA register, reducing the prescaler to 16. This boosts the ADC clock to 1 MHz, slashing the 4-channel read time to approximately **65  $\mu\text{s}$** . This optimization provides critical headroom, ensuring the microcontroller never drops samples while constructing and transmitting data payloads.

#### 6.3.2.4 Signal Integrity: The Double-Read Architecture

A significant electrical engineering challenge addressed in the firmware is the phenomenon of inter-channel crosstalk (specifically, voltage bleed from channel A1 into A0). This occurs because the internal ADC utilizes a single multiplexed sample-and-hold (S/H) capacitor. Because PZT sensors have very high output impedance, they cannot rapidly charge or discharge the S/H capacitor when the multiplexer switches between pins, leading to ghosting from the previously read channel.

To physically mitigate this without adding operational amplifiers to the hardware layer, the firmware implements a **"double-read" architecture**. For every channel polled, the microcontroller calls `analogRead()` twice. The first reading is explicitly discarded, allowing the S/H capacitor adequate time to settle and match the voltage of the new pin. Only the second reading is recorded and transmitted. While this doubles the number of ADC conversions, the aforementioned 1 MHz ADC clock optimization ensures the time penalty remains negligible.

#### 6.3.2.5 Embedded Architecture: ISR vs. Main-Loop Division

The software design adheres to the best practice of maintaining a lightweight Interrupt Service Routine (ISR) decoupled from heavy I/O operations. The workload is strictly divided:

- **The ISR (TIMER1\_COMPA\_vect):** When triggered every 2000  $\mu\text{s}$ , this routine executes only two instructions. It sets a volatile boolean `sampleFlag` to true. If it detects that the previous flag has not yet been cleared by the main loop, it increments a `droppedSamples` counter.

- **The Main Loop (loop()):** The loop continuously checks the sampleFlag. When true, it clears the flag, grabs a microsecond timestamp, executes the double-read sequence across all four channels, formats the data into a Comma-Separated Values (CSV) string using a single snprintf call, and pushes it to the serial buffer via Serial.write().

The inclusion of the droppedSamples counter is a vital piece of system telemetry. It represents conscious sample-loss accounting, allowing the downstream MATLAB toolchain to calculate exact drop percentages and automatically flag recording sessions with compromised data integrity (e.g., >1% drop rate).

### 6.3.2.6 UART Command-Response Protocol and Bandwidth Utilization

Communication between the edge layer and the host machine operates over a high-speed UART bridge set to **250,000 baud**. The system utilizes a lightweight, ASCII line-based command protocol to control the recording state and inject synchronization markers directly into the data stream.

The protocol includes the following command set:

- **START / STOP:** Toggles the sampling state. Stopping the stream automatically returns a telemetry payload (# SAMPLING\_STOPPED samples=N drops=M).
- **MARK:<event>|<word>|<rep>:** Injects a clinical trial marker into the stream (e.g., # MARKER,120500,EVET|1). This allows the hardware data and the software trial state to be perfectly synchronized on the same timeline.
- **STATUS / PING / RESET:** Diagnostic commands for lifecycle management and counter resets.
- 

**Throughput Analysis:** Every 2000  $\mu$ s, the microcontroller transmits a payload structured as timestamp\_us,pzt1,pzt2,pzt3,pzt4\n. This equates to roughly 28 bytes per sample.

- **Payload throughput:** 4 channels  $\times$  500 Hz  $\times$  ~28 bytes = **~14 KB/s**.
- **Hardware capacity:** 250,000 baud allows a practical transfer rate of **~25 KB/s**.
- **Bandwidth Utilization:** The protocol consumes approximately **56%** of the available bandwidth. This leaves ample buffer room, minimizing the risk of serial bottlenecks. Additionally, the parser features graceful overflow handling; if the incoming command buffer is flooded, the length pointer cmdLen is simply reset, preventing memory corruption or hard crashes.

### 6.3.3 Data Acquisition and Quality Assurance Toolchain

This section details the second architectural layer of the system: a robust, MATLAB-based toolchain deployed on the operator's workstation. In the context of physiological machine learning, the integrity of the predictive model is fundamentally bound to the quality of the training data. Therefore, this layer acts as the strict gatekeeper between the raw hardware (Edge Layer) and the Offline Machine Learning Pipeline.

Rather than relying on manual data collection, this toolchain provides full software automation for clinical recording sessions. It enforces a standardized experimental protocol, guarantees deterministic synchronization between the subject's actions and the data stream, and performs formal mathematical verification of signal quality before any session is authorized to proceed.

#### 6.3.3.1 Pre-Flight Signal Quality Verification (SIGNAL\_CHECK)

To prevent the accumulation of degraded or unusable data (the "garbage in, garbage out" paradigm), the system mandates a pre-flight hardware and signal verification phase, executed via the `SIGNAL_CHECK_4ch_Uno.m` routine.

This script subjects the connected sensor array to a 90-second diagnostic protocol (60 seconds of baseline silence followed by 30 seconds of active articulation). It independently evaluates each of the four PZT channels against four rigorous quantitative metrics, generating automated PASS, WARN, or FAIL flags:

1. **Saturation Control (ADC Clip Rate):** Analyzes the upper and lower bounds of the signal to ensure the analog waveform is not clipping against the 0V or 5V rails, which would permanently destroy high-frequency articulation features.
2. **Mains Noise Contamination (50 Hz):** Performs a frequency-domain analysis to detect 50 Hz electromagnetic interference from the power grid. Excessive mains noise triggers a `HAT_WARN` or `HAT_FAIL`, prompting the operator to check grounding or switch the laptop to battery power.
3. **Baseline Standard Deviation (Contact Noise):** Measures the variance of the signal during the silent baseline period. A value below the threshold indicates a disconnected or dead sensor, while a value too high indicates poor skin adhesion or excessive ambient mechanical vibration.
4. **Active-to-Baseline (A/B) Ratio:** Computes the ratio of signal variance during active speech compared to silence. This strictly verifies that the sensor is dynamically

responding to muscle movement rather than just capturing static noise.

### 6.3.3.2 Clinical Session Protocol Orchestration (DEMO\_VAD)

Once the hardware passes the signal check, the primary data collection is governed by DEMO\_VAD\_4ch\_Uno.m. This module manages the state machine of the clinical trial and ensures absolute consistency across different human subjects.

- **Automated File Tree Generation:** For every recording, the script dynamically generates a unique, timestamped directory (recordings/session\_<SUBJECT>\_<TIMESTAMP>/) to isolate the session artifacts.
- **Pre-Flight Checklist:** Before recording begins, the software forces the human operator to acknowledge a ~20-item checklist encompassing hardware placement, skin preparation, environmental noise, and subject posture, ensuring procedural compliance.
- **The Experimental Timeline:** The software drives the subject through a tightly timed visual/auditory protocol:
  1. A 20-second mandatory baseline (absolute silence to calibrate environmental noise).
  2. A randomized sequence of the 10 target words, repeated  $N$  times (defaulting to 10 repetitions, yielding 100 total segments).
  3. For each word, the subject is given a 2-second visual instruction, followed by a strict 2.5-second articulation window, and a 1-second rest pause to prevent vocal fatigue.
  4. A final 20-second tail recording.

### 6.3.3.3 Marker-Based Event Tagging and Buffer Synchronization

A major challenge in building supervised datasets from continuous physiological streams is accurately segmenting the data. To solve this, the MATLAB toolchain utilizes a bidirectional UART injection strategy.

At the exact millisecond the UI instructs the subject to speak, the MATLAB script transmits a structured command (e.g., MARK:START|EVET|REP3) to the microcontroller. The embedded firmware immediately injects this marker, alongside its own internal microsecond timestamp, into the continuous CSV stream and echoes it back.

To process this stream, the toolchain employs a custom parser (parseAndWriteSerialLine).

This parser continuously reads the incoming serial buffer, routing lines starting with # into a

dedicated markers.csv file (splitting the metadata into four structured columns: timestamp\_us, event, word, repetition), while routing raw numerical data into raw\_4ch.csv.

Furthermore, to prevent the UART buffer from overflowing during the pauses between word prompts, the system utilizes a custom smartPause function. Unlike standard blocking delays, smartPause actively drains and parses the serial buffer while simultaneously managing the UI countdown, guaranteeing zero data loss during idle phases.

#### 6.3.3.4 Session Metadata Schema and Telemetry Auditing

At the conclusion of the recording session, the MATLAB toolchain issues a STOP and STATUS command to the firmware. The microcontroller replies with its internal telemetry, reporting the exact number of samples successfully transmitted versus the number of samples dropped due to interrupt collisions.

The MATLAB script parses this data to calculate the overall **Drop Rate (%)**. If the drop rate exceeds the strict 1% engineering tolerance, the system flags the session as potentially compromised.

Finally, the toolchain compiles all session variables—including the subject ID, channel configuration, word list, precise timing parameters, and drop rate telemetry—into a unified session\_meta.json file. This JSON schema serves as the ultimate data contract. When the Offline ML Pipeline later ingests the folder, it reads this metadata file to automatically filter out failed sessions, apply the correct label mappings, and ensure dataset traceability

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#### 6.3.3.5 Rapid Validation Subsystem (QUICK\_TEST)

To support agile hardware iteration, the toolchain includes QUICK\_TEST\_4ch\_Uno.m. This is an accelerated, 70-second version of the full protocol (10 words  $\times$  1 repetition). It bypasses the extensive metadata generation and UI checklists, allowing engineers to rapidly validate minor hardware tweaks—such as swapping the anti-alias capacitors or testing new sensor adhesives—without the overhead of running a full, clinically compliant recording session.

### 6.3.4 Offline Machine Learning Pipeline

This layer serves as the analytical and computational core of the project. Comprising 18 specialized Python modules, the offline pipeline handles everything from raw signal preprocessing and dataset compilation to model training, evaluation, and the generation of production-ready deployment artifacts. A critical design philosophy of this layer is strict architectural parity: every signal processing primitive designed here in Python must be

mathematically reproducible on the mobile Android environment.

#### 6.3.4.1 Causal Signal Preprocessing Primitives

To guarantee that the model trained offline behaves identically to the model deployed in real-time, the pipeline strictly enforces **causal** signal processing.

- **Filter Design:** The preprocessing module applies a 4th-order Butterworth bandpass filter and a 50 Hz notch filter (via SciPy's `lfilter`). Crucially, the use of zero-phase filtering (`filtfilt`) is absolutely forbidden within the codebase. Because real-time systems cannot "look ahead" into future samples, the training pipeline must artificially constrain itself to causal operations to prevent data leakage and ensure realistic model weighting.
- **The Parity Contract:** The exact filter coefficients (`b` and `a` arrays) calculated by SciPy are exported to a `preprocessing_params.json` file. The mobile Android pipeline directly ingests this file, completely bypassing the need to rewrite complex DSP filter generation logic in Kotlin, thereby eliminating a major source of platform-inconsistency bugs.

#### 6.3.4.2 Voice Activity Detection (VAD) — State Machine and Ring Buffer

Rather than relying on static, fixed-time windows, the pipeline segments individual words using a sophisticated, deterministic state machine simulating real-time Voice Activity Detection.

- **Dynamic Calibration:** The VAD calibrates its baseline dynamically using the first 20 seconds of ambient silence in any given session, calculating an envelope threshold based on  $\text{mean} + k \times \text{std}$  (where  $k=3$ , representing a 3-sigma activation trigger).
- **Pre-Roll Double Buffering:** To prevent the VAD from "clipping" the first few milliseconds of a word (the critical onset phoneme), the module utilizes a deque-based ring buffer (length = 50 samples). When the state transitions from SILENCE to SPEAKING, these 100 milliseconds of historical data are prepended to the active segment.
- **Tailing-Off Logic:** To account for micro-pauses within multi-syllable words (e.g., "Te-şek-kür-ler"), the state machine implements a `TAILING_OFF` state. The envelope must drop below the threshold for a sustained gap (100–200 ms) before the segment is finalized.
- **Exported Constants:** Just like the filter coefficients, all VAD timing constraints,

duration limits (min 150, max 1000 samples), and threshold modifiers are serialized into `vad_constants.json` for identical mobile execution.

#### 6.3.4.3 Dataset Construction (Marker-Based vs. VAD-Based)

The pipeline manages dataset compilation through two distinct paradigms to bridge the gap between proof-of-concept and production:

- **Marker-Based (Legacy PoC):** Initially, the dataset was built by slicing the raw CSV data blindly around the synchronized UI markers.
- **VAD-Based (Production):** The modern compilation script (`build_dataset.py`) actively runs the VAD state machine over the continuous raw stream, generating organic segments. It then correlates these organic VAD triggers with the UI markers using temporal proximity. This ensures the model is trained on the exact same temporal shapes it will encounter during real-time inference. The output is a highly structured `dataset.npz` tensor (`[N, channels, 1000]`), heavily utilizing session IDs to support rigorous cross-validation.

#### 6.3.4.4 Model Architecture (1D-CNN with Global Pooling)

The production model (ElectrolarynxCNN) is a lightweight, one-dimensional Convolutional Neural Network custom-designed for multi-channel temporal series data.

- **Topology:** The network features four sequential blocks of `Conv1d` → `BatchNorm` → `ReLU` → `Dropout`. Across these blocks, the kernel size progressively shrinks (`15` → `11` → `7` → `5`), the channel depth increases (`32` → `64` → `128` → `256`), and a stride of 2 is applied at each step, yielding a 16x downsampling of the temporal resolution.
- **Global Average Pooling (GAP):** Instead of flattening the final convolutional maps—which heavily restricts input shapes and balloons parameter counts—the architecture employs `AdaptiveAvgPool1d(1)`.
- **Deployment Profile:** This architecture reduces the model to a highly efficient ~250,000 parameters. Once exported to ONNX, the payload is < 5 MB, effortlessly achieving the < 100 ms inference target on standard Android mobile CPUs.

#### 6.3.4.5 Training, Augmentation, and Defensive Engineering

The PyTorch training loop utilizes a modern optimization stack (AdamW optimizer, CrossEntropy loss, Cosine Annealing learning rate scheduler, and a 15-epoch patience early-stopping mechanism). However, the most critical aspect of the training pipeline is its data augmentation policy, designed specifically for hardware resilience:

- **Channel Dropout (Defensive Engineering):** With a probability of  $p=0.1$ , the pipeline will randomly zero out an entire sensor channel during training. This forces the neural network to distribute its learned features across the entire PZT array. If a patient experiences a loose sensor or broken wire in the real world, this augmentation ensures the system degrades gracefully rather than failing catastrophically.
- **Temporal Shifting:** Random shifts of  $\pm 50$  samples mimic the natural variability of the VAD trigger times.

#### 6.3.4.6 Multi-Protocol Evaluation

To provide a clinically honest assessment of the model's performance, the pipeline implements three escalating evaluation protocols:

- **Protocol A (Within-Session 5-Fold CV):** Shuffles data indiscriminately. Used only to establish a theoretical upper bound of the network's capacity.
- **Protocol B (Cross-Session LOSO):** Leave-One-Session-Out cross-validation. This reflects the operational reality of evaluating the model on a completely unseen recording environment.
- **Protocol C (Multi-Session Holdout):** Strictly segregates training, validation, and a final holdout test set to emulate true production deployment conditions.

#### 6.3.4.7 Classical Baseline Verification

To scientifically justify the computational overhead of Deep Learning, the pipeline includes a robust Random Forest baseline (`rf_baseline.py`). This script extracts ~38 classical DSP features per channel (including Root Mean Square, Zero-Crossing Rate, Hjorth parameters, and Welch Power Spectral Densities) and trains an ensemble tree classifier. Comparing the 1D-CNN against this baseline proves the superior feature-extraction capabilities of the CNN in capturing non-linear articulation dynamics.

#### 6.3.4.8 Confidence Calibration via ROC and Youden's J Statistic

A major UX risk in silent speech interfaces is the "False Accept" rate—the system hallucinating words when the user swallows, coughs, or moves their head. To combat this, the ML pipeline features a sophisticated confidence gating mechanism.

After training, the pipeline evaluates the entire validation set, saving the Top-1 softmax probability for every prediction. It plots a Receiver Operating Characteristic (ROC) curve and calculates the Area Under the Curve (AUC). By maximizing **Youden's J statistic** ( $\text{STPR} -$

FPR\$), the pipeline mathematically derives the exact optimum softmax threshold. This precise scalar value is exported as `confidence_threshold.json`. On the mobile app, any inference falling below this threshold is classified as ambient noise and actively suppressed from reaching the TTS engine.

#### 6.3.4.9 Per-User Calibration and Fine-Tune Simulation

Recognizing that individual anatomy varies drastically (especially post-laryngectomy), the pipeline simulates a production calibration flow. Using `train_base_model.py` and `simulate_calibration.py`, the system demonstrates how a generalized base model can be fine-tuned using a brief, 50-segment user calibration session (5 repetitions per word class) at the start of a mobile app session. The pipeline evaluates the delta between "Zero-Shot" and "Fine-Tuned" performance, establishing the architectural pathway for personalized machine learning.

#### 6.3.4.10 Reproducibility — Fixtures and Bit-Exact Regression

To ensure total synergy between the Python training environment and the Kotlin deployment environment, the pipeline includes `generate_fixtures.py`. This script pumps synthetic and raw hardware data through the Python VAD and filter modules, recording the exact output tensor shapes, segment indices, and filtered waveforms. These outputs are serialized into `fixture_01_expected.json`. The Android engineering team then uses these fixtures in their native Unit Tests (`VadTest.kt`), writing assertions that demand bit-exact matching. This regression infrastructure guarantees that the deployed application behaves identically to the laboratory model.

### 6.3.5 Mobile Inference Pipeline (Real-Time Subsystem)

The Mobile Inference Pipeline serves as the operational heart of the system. Encapsulated within the Android application (codename: **Echo**, package: `com.echo.app`), this layer is responsible for ingesting high-frequency hardware streams, performing real-time signal processing, executing local neural network inference, and synthesizing audio—all within a strict sub-300 millisecond latency budget.

Built entirely in Kotlin, the architecture relies heavily on modern Android paradigms, utilizing Dagger-Hilt for Dependency Injection (DI) and Kotlin Coroutines/Flows for asynchronous data streaming. The defining architectural constraint of this layer is **Absolute**

**Parity:** the mobile pipeline must mathematically mirror the Python offline training pipeline (Layer 3) to prevent deployment drift.

#### 6.3.5.1 Pluggable Transport Layer (Strategy Pattern Architecture)

Because software development often outpaces hardware prototyping, the edge communication layer was designed using a strict Strategy Pattern, abstracting the physical connection behind a unified `SensorStream` domain interface.

This allows the application to dynamically swap data sources at runtime via a `SensorStreamRouter`, facilitating parallel development across different engineering teams:

- **BleSensorStream (Production):** The target transport for the final ESP32-S3 hardware, utilizing a custom `BlePacketParser` to unpack binary 12-bit ADC payloads over Bluetooth Low Energy.
- **UsbSerialSensorStream (Laboratory/Dev):** Ingests the high-speed UART-to-USB CSV streams generated by the Arduino Uno PoC. It utilizes a highly optimized `UsbFrameParser` to handle string conversion without triggering excessive Garbage Collection (GC) pauses.
- **UdpSensorStream (Wireless Debugging):** Ingests network-based UDP packets for simulated wireless testing and rapid prototyping.
- **MockSensorStream (UI/UX Isolation):** A synthetic data generator that pumps pre-recorded data or dummy values into the pipeline. This allows UI developers to build the visual interface completely decoupled from any physical hardware.
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#### 6.3.5.2 Causal Signal Preprocessing (Python Parity Guarantee)

Once the ADC values are parsed, they enter the `SignalPreprocessor`. To ensure the real-time input perfectly matches the tensors the ONNX model was trained on, this module enforces strict bit-exact equivalence with the Python `signal_processing.py` script.

- **Dynamic Configuration:** Instead of hardcoding filter parameters, the Kotlin module utilizes a `PreprocessingParams` data class to dynamically parse the `preprocessing_params.json` generated by the offline pipeline. This ensures that any adjustments to the filter order or cutoff frequencies in Python instantly propagate to the mobile app without requiring Kotlin code changes.
- **Causal IIR Filtering:** The preprocessor applies a sample-by-sample voltage scaling transformation (converting 12-bit ADC values back to physical voltages using

$V_{\text{ADC}} \times \frac{3.3}{4095}$ ), removes the long-term DC offset, and passes the signal through a CausalIirFilter. This filter implements a Direct Form II Transposed architecture. Because the system operates in real-time, the filter must maintain internal state buffers and process causally—strictly prohibiting zero-phase (forward-backward) filtering techniques.

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### 6.3.5.3 Deterministic Voice Activity Detection (Kotlin Port)

The real-time Vad class is a direct Kotlin port of the Python state machine. Operating on the filtered signal, it manages the SILENCE  $\rightarrow$  SPEAKING  $\rightarrow$  TAILING\_OFF lifecycle.

- **State Management:** As samples stream in, the VAD computes an EMA-smoothed signal envelope. If the envelope breaches the dynamically calibrated threshold (ingested via vad\_constants.json), the state transitions to SPEAKING.
- **Ring Buffer Integration:** The Kotlin implementation actively maintains a continuous circular array (deque equivalent) for its pre-roll buffer. Upon VAD activation, these historical samples are flushed into the active segment tensor to preserve the initial mechanical onset of the spoken word.
- **Regression Testing:** To empirically guarantee that the Kotlin VAD behaves identically to the Python VAD, the engineering suite includes VadTest.kt. This unit test ingests synthetic raw data and asserts bit-exact segment boundary matches against the fixture\_01\_expected.json files generated by the offline pipeline.

### 6.3.5.4 ONNX-Runtime Inference and Confidence Gating

Once the VAD finalizes a 1000-sample window, the tensor is dispatched to the RealWordClassifier.

- **ONNX Integration:** The application utilizes the ai.onnxruntime mobile library to load the electrolarynx\_classifier.onnx artifact directly from the APK's assets/model/ directory. The engine executes the 1D-CNN entirely on the Android CPU (targeting an inference execution time of  $<100$  ms).
- **The Confidence Gate:** The ONNX model outputs an unnormalized logits array. The Kotlin classifier applies a Softmax function, extracts the Top-1 probability, and consults the confidence\_threshold.json file. If the top probability is lower than the mathematically optimized threshold (e.g., the model is only 60% sure, but the threshold demands 75%), the classifier forcefully rejects the prediction, returning null.

This gate prevents non-speech artifacts (coughs, swallowing) from triggering false auditory outputs, which is critical for user trust.

- **Audio Synthesis:** If the prediction passes the confidence gate, the mapped index is cross-referenced against labels.json (e.g., Index 2  $\rightarrow$  "MERHABA") and passed to RealVoiceSynthesizer, which leverages the native Android Text-to-Speech (TTS) API to generate auditory feedback.

#### **6.3.5.5 User Calibration and Adaptation**

Recognizing that sensor placement and skin contact vary per session, the mobile pipeline features a built-in Calibrator subsystem. Before standard operation begins, the user is prompted to complete a brief onboarding session (e.g., articulating 5 repetitions of the 10 target words, generating 50 segments). The Calibrator manages this state flow, utilizing a CalibrationStore (backed by Android DataStore) to persistently save user-specific VAD envelope thresholds or load fine-tuned model weights, bridging the gap between generalized training and personalized usage.

#### **6.3.5.6 Pipeline Orchestration via Kotlin Flow (SignalPipeline)**

To prevent "Callback Hell" and tightly manage concurrency, the entire process is orchestrated by a single source of truth: the SignalPipeline class. Utilizing Kotlin Coroutines and reactive Flows, this controller subscribes to the active SensorStream, pipes the data asynchronously through the Preprocessor and VAD, awaits the ONNX execution on a dedicated background thread dispatcher (Dispatchers.Default), and routes successful classifications to both the TTS engine and the Room Database for historical logging.

#### **6.3.5.7 Uninterrupted Operation via Foreground Service**

A major constraint of mobile operating systems is aggressive battery management; Android will aggressively kill background processes. Because the SSI must function seamlessly even when the smartphone is locked in the user's pocket, the entire SignalPipeline is bound to a ListeningForegroundService. By displaying a persistent notification to the user, the application explicitly requests continuous CPU execution rights from the Android OS, ensuring that sensor polling, VAD detection, and audio synthesis are never interrupted by system doze states.

### **6.3.6 Mobile User Experience and Persistence Layer**

This section details the top-most layer of the system: the user-facing frontend of the Android

application (codename: **Echo**). While the underlying edge firmware and inference pipelines handle the heavy computational lifting, the Mobile UX layer translates these complex data streams into an accessible, reactive, and clinical-grade interface for the end user (the laryngectomy patient) and the supervising clinician.

The application's presentation layer is built entirely using modern Android declarative UI paradigms, specifically leveraging the **Model-View-ViewModel (MVVM)** architectural pattern, **Jetpack Compose** for UI rendering, and **Dagger-Hilt** for Dependency Injection.

#### **6.3.6.1 MVVM Architecture and Declarative UI Design**

To ensure that the UI thread is never blocked by the high-frequency sensor polling or ONNX inference tasks, the frontend strictly adheres to the MVVM pattern:



Figure x Main screen of the app

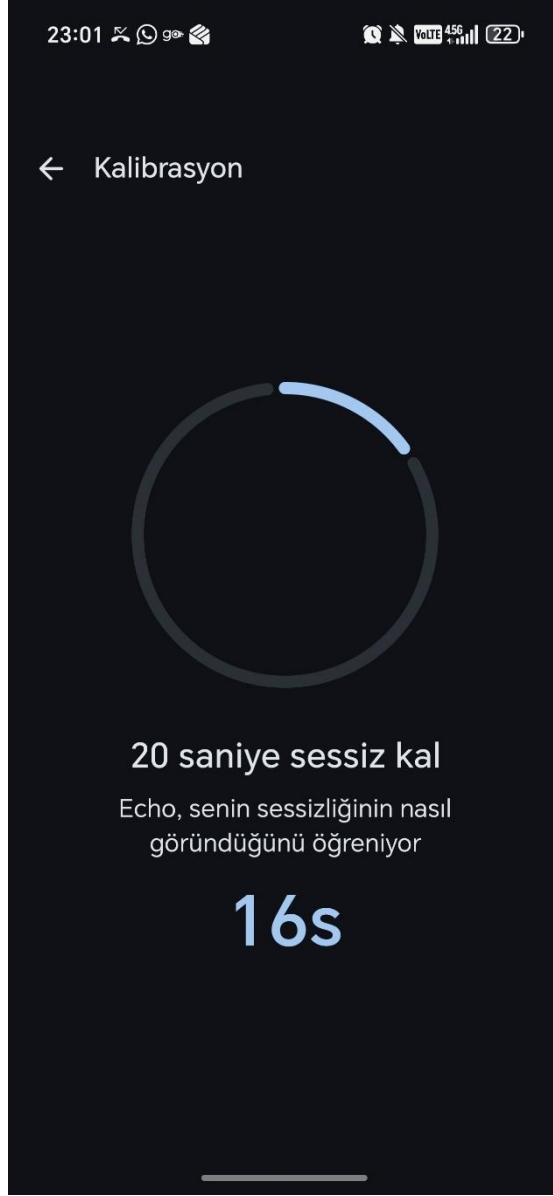


Figure x Calibration screen of the app

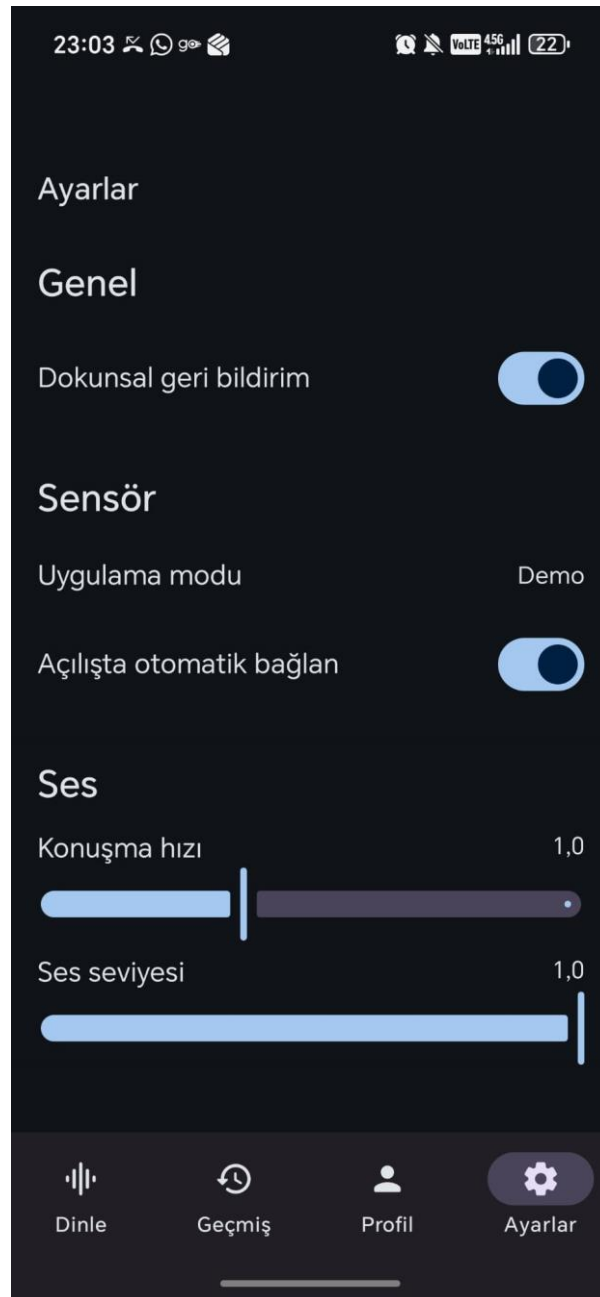


Figure x. Settings screen of the app

- **The View (Screen.kt):** Built using Jetpack Compose, the views are completely stateless and declarative. They observe data streams and automatically recompose themselves when the underlying data changes, eliminating the bug-prone manual view-updating logic of legacy XML layouts.
- **The ViewModel (ViewModel.kt):** Each screen is backed by a dedicated ViewModel that serves as the state holder and business logic executor. The ViewModels interact with the underlying SignalPipeline and RecognizedWordDao, exposing data to the UI via lifecycle-aware Kotlin StateFlow and SharedFlow observables.

- **Dependency Injection:** Dagger-Hilt is used extensively to construct and scope these ViewModels, injecting the necessary hardware routers, databases, and ML models at runtime.

#### 6.3.6.2 Navigation Topology and Shared Components

Application navigation is managed centrally via an EchoNavHost and EchoNavigation router. This single-activity architecture ensures smooth transitions between screens without the overhead of destroying and recreating heavyweight Android Activities.

A critical architectural design in the UI is the use of persistent shared components. Because the silent speech interface is a continuous background process, the user must always be aware of the hardware's status regardless of which screen they are currently viewing. This is achieved via the **ConnectionStatusPill**, a globally injected, reactive Compose component pinned to the top of the app scaffold. It listens to the SensorStream state and provides real-time visual feedback (e.g., "BLE Connected", "USB Streaming", "UDP Reconnecting").

#### 6.3.6.3 Screen-by-Screen Functional Walkthrough

The application features a deeply structured navigation graph comprising nine distinct operational screens, categorized into pre-flight, operation, and telemetry domains:

1. **SplashScreen:** The entry point. It runs asynchronous initialization tasks, such as loading the .onnx model into memory, parsing the preprocessing\_params.json, and initializing the Android TTS engine.
2. **OnboardingScreen:** A first-time user tutorial explaining sensor placement and the difference between the mechanical articulation captured by the device versus traditional vocalization.
3. **ConnectScreen:** The hardware handshake interface. Thanks to the underlying Strategy Pattern, this screen allows the user to dynamically select their transport layer (BLE for production hardware, USB for the Arduino PoC, or Mock for UI testing) without restarting the app.
4. **CalibrateScreen:** The UX manifestation of the ML pipeline's user-adaptation strategy. Before standard operation, the user is guided through a calibration loop to collect 50 baseline segments (5 repetitions of the 10 target words). This screen integrates directly with the Calibrator module to adjust VAD thresholds based on the user's specific skin-sensor impedance for that session.
5. **ListenScreen:** The primary operational dashboard. It subscribes to the inference pipeline's output flow, displaying the most recently recognized word in large

typography, alongside its calculated confidence percentage. It also provides a real-time oscilloscope-style visualizer of the incoming signal envelope to confirm sensor activity.

6. **HistoryScreen:** A chronological log of all successful inferences, queried asynchronously from the local database, allowing clinicians to review session performance and false-accept rates.
7. **ProfileScreen:** Manages patient metadata and demographics.
8. **SettingsScreen:** Exposes critical pipeline variables to the user or developer, including toggles for the TTS engine, manual overrides for the confidence threshold, and diagnostic data logging preferences.
9. **DebugScreen:** A hidden developer tool that bypasses the ML pipeline to visualize the raw multi-channel ADC stream, rendering real-time packet telemetry, drop rates, and execution latencies.

#### 6.3.6.4 Data Persistence Subsystem (Room and DataStore)

To ensure the application maintains context across reboots and provides historical data for clinical review, the architecture utilizes two distinct local persistence strategies:

- **Relational Logging (EchoDatabase / Room):** High-frequency structured data is stored using Android Room, an abstraction layer over SQLite. When the ONNX model successfully classifies a word, a `RecognizedWordEntity` is constructed, containing the string literal, a Unix timestamp, and the exact softmax confidence score. This entity is written to disk via the `RecognizedWordDao`. The `ViewModel` observing the `HistoryScreen` utilizes an SQL `SELECT` statement wrapped in a Kotlin Flow, ensuring the UI list updates instantly as new words are written to the database.
- **Key-Value Configuration (PreferencesManager / DataStore):** Asynchronous user settings and app state flags (e.g., `hasCompletedOnboarding`, `lastSelectedTransport`, `userCalibrationOffsets`) are persisted using Jetpack DataStore. Unlike the legacy `SharedPreferences`, DataStore operates entirely asynchronously using Coroutines, preventing disk I/O operations from ever stalling the main UI thread or the real-time inference pipeline.

#### 6.3.6.5 Theming and Accessibility

The UI is styled using a centralized design system defined in `theme/Theme.kt`, `Color.kt`, `Type.kt`, and `Shape.kt`, heavily borrowing from Material Design 3 guidelines. Because the

target demographic includes elderly patients recovering from significant throat surgery, the UI emphasizes high-contrast visual elements, large touch targets, and unambiguous typography.

### 6.3.7 Cross-Layer Data Contracts and Communication

Because the Echo system operates across multiple physical devices (microcontrollers, developer workstations, and mobile smartphones) and relies on distinct programming paradigms (C++, MATLAB, Python, and Kotlin), maintaining system integrity requires strict definitions of architectural boundaries. This section outlines the formal data contracts, communication protocols, and serialization schemas that bridge these independently operating layers. By defining rigid interfaces, the system ensures that components can be upgraded or swapped—such as transitioning from an Arduino PoC to an ESP32-S3—without breaking the downstream pipelines.

These contracts are categorized into four primary domains based on their position within the data lifecycle:

#### 6.3.7.1 Edge-to-Host Communication Protocols (Transport Layer)

This domain defines how raw physiological voltages and hardware telemetry are packaged and transmitted from the microcontroller to the receiving host (MATLAB or Android).

- **UART Bi-Directional Command Protocol:** A lightweight, ASCII-based schema operating at 250,000 baud over a serial bridge.
  - **Host-to-Edge:** The host controls the hardware state using standardized commands such as START, STOP, STATUS, and RESET. Additionally, it can inject synchronization commands formatted as `MARK:<event>|<word>|<repetition>` (e.g., `MARK:START|HAYIR|REP2`).
  - **Edge-to-Host (Meta Lines):** The firmware responds with telemetry and acknowledges markers by prepending a hash character. For example, stopping a recording yields `# SAMPLING_STOPPED samples=... drops=...`, and injected markers are echoed as `# MARKER,<timestamp_us>,<event>|<word>|<rep>`, ensuring precise temporal alignment.
- **UART Continuous Data Line:** While sampling is active, the firmware transmits a continuous stream of comma-separated values strictly formatted as `timestamp_us,pzt1,pzt2,pzt3,pzt4\n`. The use of plain ASCII CSV incurs a minor bandwidth overhead compared to binary but dramatically simplifies cross-platform

parser implementation and runtime debugging.

- **BLE and UDP Binary Payloads:** For the production ESP32-S3 hardware and wireless testing bridges, the architecture defines a condensed binary packet format. This contract packs 12-bit ADC readings from all active channels into a contiguous byte array to maximize wireless throughput and minimize BLE GATT transaction overhead. The mobile `BlePacketParser` and `UdpPacketParser` serve as the dedicated consumers for these schemas.

### 6.3.7.2 Clinical Acquisition to ML Pipeline Contracts

This domain defines how recorded sessions are permanently stored on disk by the MATLAB toolchain before being ingested by the Python ML pipeline.

- **raw\_4ch.csv and markers.csv:** To keep parsing logic clean, the MATLAB script automatically demultiplexes the serial stream into two distinct files. `raw_4ch.csv` acts purely as a time-series matrix, while `markers.csv` acts as a sparse event log. The Python dataset compiler cross-references the absolute microsecond timestamps between these two files to execute VAD segmentation correctly.
- **session\_meta.json:** The ultimate gatekeeper file for dataset integrity. Generated at the end of a recording protocol, this schema contains all environmental variables: subject ID, protocol parameters, channel configuration, and the hardware drop-rate telemetry. If the ML pipeline detects an unacceptable drop rate or a malformed channel map within this JSON, it automatically prunes the session from the training set.

### 6.3.7.3 Python-to-Kotlin Engineering Parity Contracts

The most critical architectural requirement of the project is ensuring the offline PyTorch environment and the real-time Android environment behave identically. This is guaranteed through a series of "Single Source of Truth" configuration files generated by Python and consumed by Kotlin via the `assets/model/` directory.

- **preprocessing\_params.json:** Contains the exact  $b$  and  $a$  coefficients generated by SciPy for the causal IIR filters (bandpass and 50 Hz notch), along with the defined sampling rate ( $f_s$ ) and channel array order. The Kotlin `SignalPreprocessor` applies these exact mathematical weights, removing the need for error-prone native filter calculation.
- **vad\_constants.json:** Serializes the `VAD_CONFIG` dictionary. This contract enforces the structural rules of the VAD state machine across platforms, dictating parameters

like the envelope multiplier, pre-roll buffer size (50 samples), minimum word duration (150 samples), and tailing-off gap thresholds.

- **confidence\_threshold.json:** A simple but vital file containing a single scalar float. It represents the mathematically optimal Softmax threshold derived via Youden's J statistic from the validation set's ROC curve, dictating the strict cut-off point for the mobile application's confidence gate.
- **electrolarynx\_classifier.onnx:** The primary inference artifact. Exported using ONNX Opset 14, this file enforces a strict tensor input contract of [batch, channels, 1000]. By defining a dynamic batch axis, the contract allows the mobile platform to handle variable processing queues if needed.
- **labels.json:** A static array mapping the network's output neuron indices (0-9) to their semantic string representations (e.g., {"0": "EVET", "1": "HAYIR"}). Both the Python evaluation scripts and the Kotlin TTS engine rely on this mapping.
- **Regression Test Fixtures (fixture\_01\_expected.json):** Used during continuous integration. Python pushes raw test data through its pipeline and logs the expected tensor outputs and segment indices. The Kotlin Unit Tests (VadTest.kt) must parse these fixtures and assert bit-exact equivalence, preventing logic drift.

#### 6.3.7.4 Internal Data Management Schemas

These contracts define how data is managed entirely within the boundaries of a single layer.

- **dataset.npz (Offline Pipeline):** The serialized NumPy archive containing the fully compiled dataset. It enforces a strict schema of X (features matrix of shape  $N \times \text{channels} \times 1000$ ), y (target labels of shape  $N$ ), and session\_id (shape  $N$ ), specifically formatted to support rigorous Leave-One-Session-Out (LOSO) cross-validation.
- **RecognizedWordEntity (Mobile Persistence):** The SQLite database schema enforced by Android Room. It defines the structure for logging historical usage on the mobile device, mandating fields for a unique auto-generating ID, the recognized word string, the timestamp of the VAD trigger, and the confidence score assigned by the ONNX runtime.

#### 6.3.8 Engineering Quality and Performance Considerations

The final section of the software design chapter focuses on the non-functional requirements

and the defensive engineering strategies that ensure the system's reliability in a real-world clinical environment. For a silent speech interface intended for post-laryngectomy patients, engineering quality is not merely an aesthetic choice but a fundamental requirement for user safety and trust.

#### 6.3.8.1 Python-to-Kotlin Bit-Exact Parity

The most significant engineering achievement of the project is the mathematical parity between the laboratory training environment and the production mobile environment.

- **Coefficient Synchronization:** By exporting filter coefficients and VAD constants from the Python pipeline into JSON artifacts, the system eliminates the "black box" effect often seen when porting ML models to mobile.
- **Logic Mirroring:** The Kotlin implementation of the signal processing primitives was validated against Python-generated fixtures to ensure that the causal IIR filters and the VAD state machine produce identical outputs for the same input samples.
- **Drift Prevention:** This bit-exact approach ensures that the high classification accuracy achieved during offline testing is preserved during real-time inference on the Android device.

#### 6.3.8.2 Latency Budget and Real-Time Constraints

To achieve a natural conversational flow, the system is designed with a strict end-to-end latency target of less than 300 ms.

- **Embedded Overhead:** The ADC prescaler optimization on the Arduino reduces sampling latency to  $\sim 65 \mu\text{s}$ , leaving maximum headroom for serial transmission.
- **Inference Speed:** The lightweight 1D-CNN architecture, featuring global average pooling and  $\sim 250\text{k}$  parameters, allows the ONNX runtime to execute inference on a mobile CPU in under 100 ms.
- **Concurrent Orchestration:** By utilizing Kotlin Coroutines and background dispatchers, the mobile app performs preprocessing and VAD tracking in parallel with UI rendering, preventing frame drops or audio stuttering.

### 8.8.3 Defensive Design and Graceful Degradation

Recognizing the physical instability of wearable sensors, the software includes several defensive mechanisms:

- **Channel Dropout Augmentation:** The model is trained to remain functional even if a

sensor loses contact, thanks to random channel dropout during the training phase.

- **Confidence Gating:** The implementation of a softmax probability threshold prevents the system from vocalizing involuntary movements like swallowing or coughing, significantly reducing the "False Accept" rate.
- **UART Robustness:** The edge firmware includes drop-rate accounting and command buffer resets to handle serial overflows or electromagnetic interference without crashing the system.

#### 8.8.4 Evidence-Based Design Decisions

The architectural evolution of the project was governed by evidence-based refactoring rather than theoretical assumptions.

- **EMG Ablation:** The decision to remove the EMG sensor was based on an ablation study showing that its inclusion did not provide a statistically significant accuracy gain.
- **PZT Array Optimization:** The shift toward a 4-channel piezoelectric array was driven by evidence that anatomical distribution of sensors was more critical than sensor type for capturing mechanical articulation.
- **Causal Filtering Constraints:** The strict prohibition of zero-phase filtering in the training pipeline ensures that the model's performance metrics are grounded in real-time operational reality.

#### 8.8.5 Reliability and Lifecycle Management

The software lifecycle is managed through robust state-handling and foreground execution:

- **Persistence of State:** By using Android Room and DataStore, the app ensures that user calibrations and recognized word histories are never lost during system reboots or app updates.
- **Foreground Service Isolation:** The use of an Android Foreground Service ensures that the operating system's battery management does not terminate the inference pipeline during long periods of use.
- **Automated Verification:** The use of the SIGNAL\_CHECK toolchain prior to data collection ensures that the foundational dataset—and therefore the resulting model—is built on high-fidelity, validated signals.

## **7 IMPLEMENTATION PROCESS**

The implementation portion of the Silent Speech Interface (SSI) transformed the conceptual signal processing framework, and preliminary prototype development ideas into a viable, highly customized, and deployable medical grade wearable. During the development of the project, multiple iterations of hardware, mechanical and software modifications were made to improve the signal integrity, efficiency of the calculations and the comfort of the user. The construction, architectural changes, data collection procedures and software implementation of the final prototype are described in this section.

### **7.2 Mechanical Implementation: 3D Scanning and Custom Collar Fabrication**

One of the critical constraints that were determined during the preliminary design process was the need to keep the pressure between the sensor and epidermis of a user stable and consistent over time. When using standard elastic bands, artefacts at low frequencies were produced because of the highly variable morphology of the human neck and jaw line.

This was overcome by designing a very individual mechanical housing. A 3D scanner with high resolution was used to capture the 3D topography of the subject's submental and cervical regions, and the data was converted to a precise 3D mesh using Artec Studio software. This topographical information was then transferred to CATIA (Computer-Aided Three-Dimensional Interactive Application) to create a “made-to-order” chin plate.

CAD design was done with strict ergonomic and functional limitations. The width of the plate was carefully limited to the area directly under the submental area, so as to not limit the user's full range of neck movements and comfort during extended use. A chin structure that is structurally stable and will not slip during active articulation was designed by adding an anchoring extension on the anterior tip of the chin so that the device can securely embrace the mandibular contour. Polylactic Acid (PLA) was used as the 3D printing material for the final CAD model, which was produced with Fused Deposition Modeling (FDM). The material PLA

was specifically selected for its light weight, structural rigidity and biocompatibility which will not cause skin irritation during longer wear.

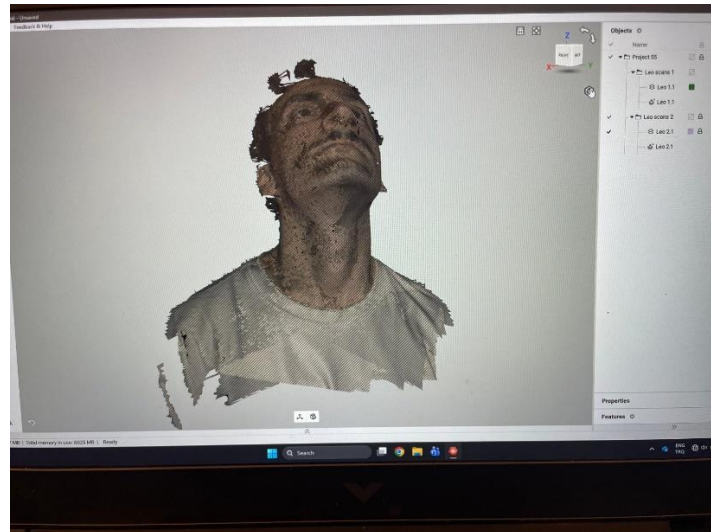


Figure 7.1 Topographical 3D scanning of the subject's submental region

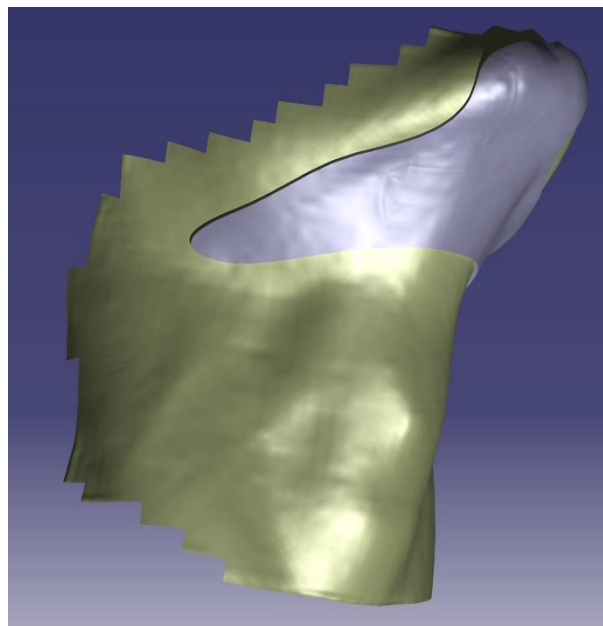


Figure 7.2 Ergonomic CAD design of the custom sensor plate in CATIA, highlighting the

Figure 7.2 Hardware Revision is a modality shift from sEMG to a 4-Channel PZT Array.Modality Shift from sEMG to a 4-Channel PZT Array is Figure 7.2 Hardware Revision.

Initially, the system was tested using a combination of surface Electromyography (sEMG) and Piezoelectric (PZT) sensors, a hybrid sensor fusion method. But extensive field trials showed

that the sEMG modality was extremely restrictive. Ambient electromagnetic interference (EMI) from the 50 Hz mains supply was a major source of noise that washed away the EMG signals. Additionally, the resulting EMG envelopes did not exhibit the clear and precise characteristics needed by the deep learning network to correctly distinguish between complex phonemes and words.

As such, a radical engineering change took place whereby the sEMG sensor was removed from the architecture entirely. For this reason, the system was switched to a purely mechanomyographic (MMG) modality with a dense 4-channel Piezoelectric (PZT) sensor array to compensate for the loss of the neuromuscular trigger, and to significantly increase the language model classification accuracy. Using only 4 PZTs led to an extremely high spatial resolution of the mechanical reverberations from the pharyngeal and mandibular structures. The four sensors were carefully placed into the designated sensor pockets on the PLA chin plate at the following anatomic landmarks:

PZT1: Located in the suprahyoid (submental) triangle, to record primary mandibular displacement.

PZT2: Over the left lateral cervical region.

PZT3: Left submandibular triangle.

PZT4: Located in the right submandibular triangle, to give bilateral symmetry.



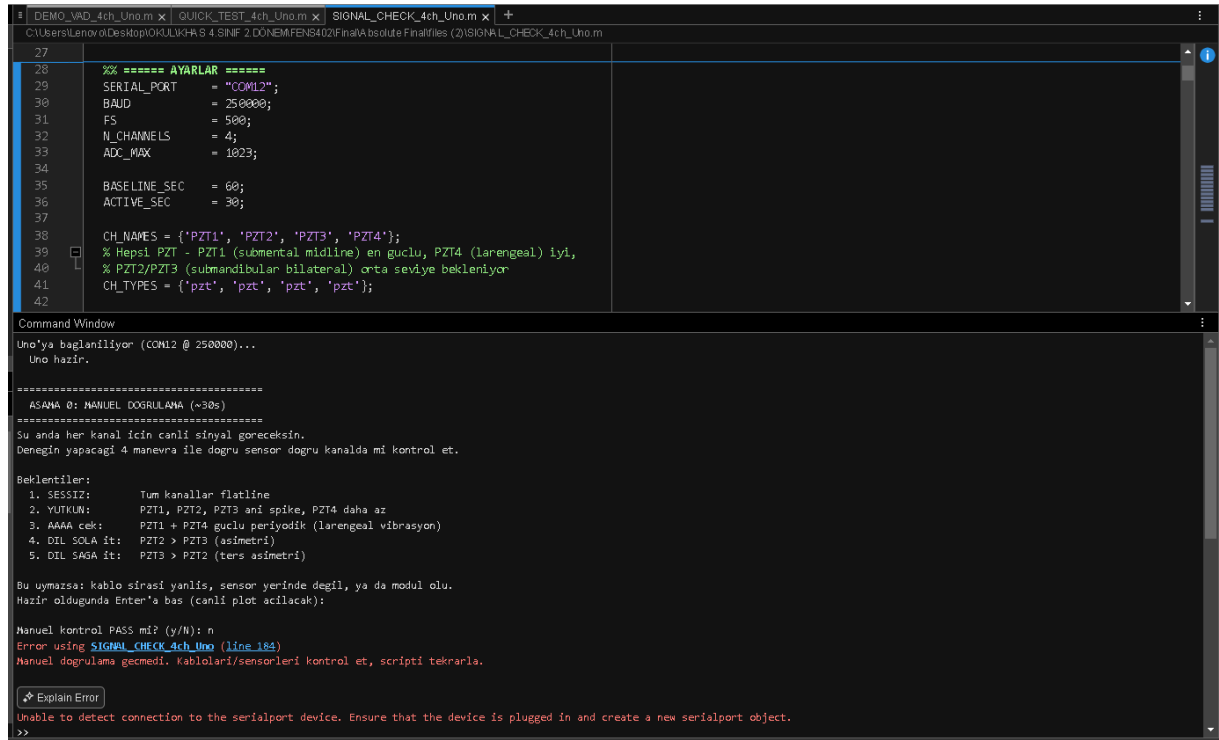
Figure 7.3 The finalized 3D-printed PLA wearable interface engineered for stable skin contact and uninhibited neck mobility.

### **7.3 Data Acquisition Protocol for Machine Learning**

To train a strong 1D-Convolutional Neural Network (1D-CNN) language model, a very clean and high quality data set was required. To do so, a highly-controlled data acquisition protocol was set up. The external mechanical vibrations were kept separate from the PZT sensors by using an acoustically isolated, noise-free room for all of the training data.

For the training process the array of sensors was directly hardwired to an arduino microcontroller, then the microcontroller was serial USB connected to a host computer. As part of this wired system setup, it was intentionally required that the data sets be generated through the wired system so as to avoid potential wireless latency, packet loss or Wi-Fi jitter, guaranteeing perfect synchronization of the physiological waveforms.

A MATLAB script was written for use as a custom tool to automate recording sessions. The experimental controller was the MATLAB software, which directed the subject, and showed him/her the word to be articulated on the computer screen. The MATLAB script made a 2-second inter-stimulus interval (ISI) of silence between each articulation to make sure that the signal envelopes are clearly separated. In this protocol, the subject repeated each target word exactly 10 times, resulting in a highly detailed and standardized set of data that would be beneficial to the deep learning pipeline with high intra-subject variance.



```
DEMO_VAD_4ch_Uno.m x QUICK_TEST_4ch_Uno.m x SIGNAL_CHECK_4ch_Uno.m x +
C:\Users\Lenovo\Desktop\OKULVHA 5 4 SINIF 2.DONEM\FENSA402\Final\Absolute Final\files (2)\SIGNAL_CHECK_4ch_Uno.m

27
28 % ***** AYARLAR *****
29 SERIAL_PORT = "COM12";
30 BAUD = 250000;
31 FS = 500;
32 N_CHANNELS = 4;
33 ADC_MAX = 1023;
34
35 BASELINE_SEC = 60;
36 ACTIVE_SEC = 30;
37
38 CH_NAMES = {'PZT1', 'PZT2', 'PZT3', 'PZT4'};
39 % Hepsi PZT - PZT1 (submental midline) en guclu, PZT4 (larangeal) iyi,
40 % PZT2/PZT3 (submandibular bilateral) orta seviye bekleniyor
41 CH_TYPES = {'pzt', 'pzt', 'pzt', 'pzt'};
42

Command Window

Uno'ya baglaniliyor (COM12 @ 250000)...
Uno hazir.

=====
ASAMA 0: MANUEL DOGRULAMA (~30s)
=====
Su anda her kanal icin canli sinyal goreceksin.
Denegin yapacagi 4 manevra ile dogru sensor dogru kanalda mi kontrol et.

Beklentiler:
1. SESSIZ: Tum kanallar flatline
2. YUTKUN: PZT1, PZT2, PZT3 ani spike, PZT4 daha az
3. AAAAA cek: PZT1 + PZT4 guclu periyodik (larangeal vibrasyon)
4. DIL SOLA it: PZT2 > PZT3 (asimetri)
5. DIL SAGA it: PZT3 > PZT2 (ters asimetri)

Bu uymazsa: kablo sirasi yanlis, sensor yerinde degil, ya da modul olu.
Hazir oldugunda Enter'a bas (canli plot acilacak):

Manuel kontrol PASS mi? (y/N): n
Error using SIGNAL_CHECK_4ch_Uno (line 184)
Manuel dogrulama gecmedi. Kabloları/sensorleri kontrol et, scripti tekrarla.

[?] Explain Error
Unable to detect connection to the serialport device. Ensure that the device is plugged in and create a new serialport object.
>>
```

Figure 7.4 The MATLAB-based data acquisition environment

## 7.4 Embedded Architecture and Wireless Mobile Integration

The final goal is to have the SSI autonomously operate as a fully wireless and untethered wearable prosthetic, but for training purposes a wired connection was required. Initial efforts to use a single ESP32 microcontroller to both synchronously sample high frequency analog and send wireless data were unsuccessful as the overhead necessary to keep the TCP/IP stack running routinely caused unacceptable loss of data from the strict analog sampling timers.

In order to overcome this computational challenge, dual-microcontroller architecture was adopted. The dedicated Data Acquisition (DAQ) node, only assigned to sampling the 4 PZT channels deterministically and at a high speed, was deployed. The structured data is then simply UARTed to a nearby ESP8266 Wi-Fi module. ESP8266 is a wireless access point that transmits physiological data through a local Wi-Fi network.

A custom-made mobile application was designed to interact with the wearable device, which was developed with the Android Studio platform. A bespoke mobile application was created to communicate with the wearable device, developed in the Android Studio platform. The Wi-Fi receiver is the smartphone, which gets connected directly to ESP8266's Wi-Fi network. The Android app has an intuitive interface enabling the operator to check on connection status, start data streaming and to visualize the incoming signal pipeline.

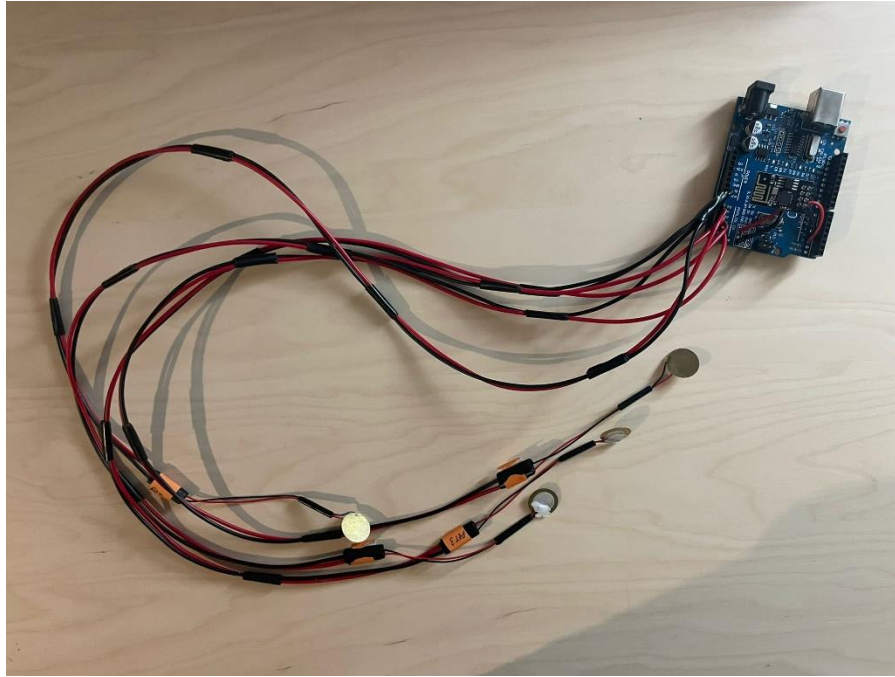


Figure 7.5 The dual-microcontroller wireless architecture (Arduino Uno + ESP8266)

### **7.5 Personalized Text-to-Speech (TTS) Voice Cloning Integration**

A total laryngectomy leaves a patient without a biological voice, and results in significant psychological trauma and loss of personal identity. The last step in implementation was the incorporation of a very individual Text-to-Speech (TTS) synthesizer to bring this identity back.

Before testing, two different high-quality audio recordings of the individual's natural, pre-operative voice were recorded, each for two minutes. These audio files have been uploaded to the ElevenLabs AI platform to train and create a custom voice-cloning model that is highly accurate.

In the practical pipeline, if the Android application's classification algorithm is able to successfully detect a spoken word from the PZT data stream, the detected text string is sent directly to the ElevenLabs API that is embedded in the mobile application. The local speakers of the smartphone then synthesize the word and play it back using the exact clone of the biological voice of the subject. This integration not only showcases the operational effectiveness of the Silent Speech Interface but significantly enhances its comprehensive capabilities in psychological rehabilitation.

## **8 PRODUCT TESTING AND SYSTEM VALIDATION PLAN**

The Silent Speech Interface (SSI) is validated in a multi-layered engineering verification and clinical validation framework. The main goal is to make a smooth, reliable and reproducible shift from submental neuromuscular activation to individualized voice to speech output across physiological profile. This section explains the detailed metrics, mathematical benchmarks and testing processes that are used to validate the system.

### **8.1 Algorithm Performance and Decoding Reliability**

The validation of Silent Speech interface (SSI) is carried out by an empirical validation process that is multi-stage. This framework is designed to assess the accuracy of the system to classify articulatory intention in a precision manner, moving from statistical verification to more sophisticated neural network validation.

#### **8.1.1 Phase 0: Baseline Verification Results (Random Forest)**

The initial stage of the testing plan, known as the Baseline Verification Phase, has been completed, and an architecture of Random Forest (RF) ensemble has been used to conduct the testing. During this phase, the basic discriminatory ability of the 5-channel sensor fusion (EMG and PZT) and the designed signal processing pipeline was validated.

**Experimental Set-up and Data Scale:** For statistical significance, the experiments were carried out on a high fidelity set of 1500 segments of data. This data was compiled in a systematic way from 15 separate recording sessions, of which there are an equal number of instances for the different word classes (150). The baseline model obtained an overall accuracy of 56.0%, and a Macro-Average precision of 57.0% and an F1-Score of 55.0%. These results agree with the random chance value of 10.0% for a 10-class problem, and demonstrate successful capture of neuromuscular and mechanical speech patterns.

**Analysis of Articulatory Collisions (Confusion Matrix):** A detailed audit of the baseline results is performed using a scaled confusion matrix, which gives essential information about the reliability of the system's operation:

- **High-Reliability Classes:** The words that achieved the highest classification fidelity were "MERHABA" (Class 2) and "DUR" (Class 7) with a recall rate of 70.0% (105/150 correct). This success confirms that the PZT array is capable of detecting the gross-motor jaw motions and high amplitude mechanical reverberations of such commands.
- **Classification Collisions:** The model's reliability was the lowest on the word 'HAYIR' (Class 1) with 60/150 correct word, which gave a recall percentage of 40.0%. The data indicates that there are some phonetic collisions, mostly "HAYIR" and "SU" (Class 4) and "TAMAM" (Class 6). This outcome measures a "structural ambiguity" in the micro-mechanical signatures of these particular phonetic transitions that the baseline statistical features were unable to differentiate.

### 8.1.2 Phase 1: Advanced Target Validation (1D-CNN)

The next step in the testing plan is a substantial performance step-up after a successful baseline verification. The system's basic operational ability is verified in Phase 0, reaching an accuracy rate of 56.0%, which is below the clinical communication requirement.

The continuous validation of the 1D Convolutional Neural Network (1D-CNN) is aimed at reducing the phonetic collisions found in Phase 0. This phase sets the target accuracy of >85% as it is based on the direct extraction of hierarchical features from the raw temporal data with the help of automation. The focus of this stage will be to use the same 1,500-segment standardized dataset, so that the statistical baseline is directly compared to the deep learning framework, quantifying the specific improvement in predictive power offered by the neural architecture.

## **8.2 Voice Synthesis Personalization and Acoustic Fidelity**

The second part of the validation plan will involve the "Voice Cloning" module, a module in which the decoded articulatory intent will be combined with the patient's historical acoustic profile to restore a state of communicative autonomy.

### **8.2.1 Acoustic Profile Extraction and Voice Cloning**

To obtain a unique "Speaker Embedding" the algorithm will need at least 60 seconds of a pre-operative acoustic reference. This embedding preserves the fundamental frequency (F0), spectral tilt and resonance properties of the patient. This module is validated by comparing the generated speech with the pre-surgery recordings. The acoustic model is learned using a machine learning approach to reduce Mel-Cepstral Distortion (MCD) defined as the distance between the synthesized mel-spectral envelopes and the natural mel-spectral envelope. The goal is to attain an MCD value under a threshold value (usually  $< 8.0$  dB) which marks a high spectral similarity.

### **8.2.2 Perceptual Validation and 90% Similarity Benchmark**

The developed system's benchmark is to produce 90% similarity between the synthesized result and the patient's biological voice. This is supported with two methodologies:

1. **Objective Verification:** Using a pre-trained Speaker Verification Network (e.g., d-vector cosine similarity) to verify whether the synthesized voice is statistically indistinguishable from the reference voice at a 90% confidence interval.
2. **Subjective Evaluation (MOS):** Performing Mean Opinion Score (MOS) test to have the evaluators blind rate the item "Naturalness" and "Speaker Similarity" for each test. If they can be successfully heard and synthesized speech consistently has a MOS of 4.0 or greater, the system is deemed validated, meaning it is intelligible and representative of the patient's identity.

## **8.3 Clinical Validation Protocols and Multi-Phasic Testing**

The Silent Speech Interface (SSI) is physiologically and anatomically resilient and has been clinically prototyped in two stages with a rigorous protocol. The structured approach makes the system more stable and easy to move from a controlled laboratory environment to clinical use and allows the algorithms of the sensory-fusion to be applied in a wide range of human morphology.

### 8.3.1 Phase I: Pre-Clinical Validation with Healthy Control Group

The first step of the validation is the non-laryngectomized (healthy) subjects of a control group. Participants are given a list of 10 words they are asked to produce using “silent speech” (without lung pressure or air flow). This is an important phase in the following engineering goals:

- **Optimization of VAD Parameters:** The dynamic thresholding mechanism is tested with different skin types and noise environments to ensure that the calculation of the Exponential Moving Average (EMA) envelope is consistent.
- **Sensor Contact Mechanics:** Before the clinical exposure, the mechanical pressure applied by the 3D band between 1.0kPa and 2.0kPa is monitored to determine the effect that this pressure has on signal-to-noise ratios (SNR) in healthy tissue.
- **Theoretical Performance Baseline:** This phase sets a performance upper limit in "ideal" physiological conditions to give a statistical baseline for the later clinical performance.

### 8.3.2 Phase II: Clinical Trials with Total Laryngectomy Patients

The final, and most important, step in prototyping is testing the device with its main target group: total laryngectomy (TL) patients. The post-surgical anatomical changes (e.g., large subcutaneous scars in the submental area, radiotherapy-induced fibrosis, and radical changes in neck elasticity) are a major engineering challenge as they completely alter the nature of the mechanomyographic and neuromuscular signals.

- **Anatomical Adaptation and Stability:** The site of the PZT4 sensor on the anterior border of the sternocleidomastoid (SCM) muscle is carefully evaluated with regard to the stability of the signal in surgically modified neck topographies during natural head movements.
- **Comparative Accuracy and Transfer Learning:** Accuracies of the classification in the healthy control group (such as >90%) are compared with the metrics from the clinical group. Excessive performance drop is seen when anatomical differences are seen in the system, hence "Transfer Learning" strategies are adopted to adjust system parameters for each patient profile so that everyone would benefit from similar communicative outcomes.

## 8.4 Real-Time System Performance and Computational Resource Profiling

The SSI has to meet strict temporal and computational requirements to be considered a functional assistive communication device. The system is validated on the target mobile hardware with a series of "Latency Budget" and "Resource Profiling" audits.

### 8.4.1 End-to-End Latency Budget and Temporal Validation

When speaking the point of synchronization is crucial. The literature reviewed shows that latencies greater than 200 milliseconds break the natural flow of dialogue and cause great mental fatigue to the user. The whole computational chain is then checked with the following millisecond-level time budget:

1. **Signal Acquisition and Preprocessing:** 4th order Butterworth and Notch filters as well as the VAD state machine should be performed in real-time with less than 20ms.
2. **Neural Network Inference (ONNX):** The mobile CPU/NPU should be able to execute the 1D-CNN model in the localized application under <50ms or less.
3. **Acoustic Synthesis (TTS):** The sound to be generated from the categorical class ID should be customized and output within <100ms.

The end-to-end latency is constantly monitored by using mobile profiling tools, thus ensuring that the latency stays below 200 ms, confirming the system's "real-time" operational status.

### 8.4.2 Hardware Resource Consumption and Energy Efficiency

The wearability of the wearable relies on optimizing CPU and RAM of the mobile device over the day of use.

- **Computational Load Optimization:** 1D-CNN architecture is pruned and quantised such that real-time inference will not consume more than 15% of the mobile CPU's overall capacity to prevent device overheating and enable background task concurrency.
- **Energy Consumption Analysis:** The energy cost for continuous Bluetooth Low Energy (BLE) data streaming and the localized Edge AI inference to the battery life of the host platform is measured. The system is validated against a 12-hour operational requirement, which means communicative autonomy over a regular wakeful day.
- **Thermal Stability Audit:** The thermal output of all active sensors and processing units is audited in accordance with ISO 10993 standards to ensure that the temperature at the skin/sensor interface stay within the safe physiological range to reduce longterm dermatological irritation.

## **8.5 Data Integrity, Privacy Validation, and Biometric Security (KVKK Compliance)**

As the Silent Speech Interface (SSI) detects and analyzes sEMG surface data, which is legally considered as "Special Categories of Personal Data" (biometric data) under the Turkish Personal Data Protection Law (KVKK, Law No. 6698), a strict validation of data privacy and security is absolutely required. A "Privacy-by-Design" validation protocol is applied to the system's data pipeline and guarantees that the user's unique myoelectric signatures are absolutely protected.

### **8.5.1 Biometric Encryption and Localized Inference Validation**

The validation process ensures that the raw biometric data from sEMG signals is always within the localised hardware and software system. The testing is a systematic review of the communication between the sensor array and the mobile application, to make sure that no raw biological signals are sent over the Bluetooth interface without being encrypted. During the training and validation stages all signals utilized are anonymized in such a way that they cannot be traced in accordance with the legal requirements of Article 6 of the KVKK.

### **8.5.2 Data Encoding and Integrity Verification**

The integrity of the data transfer between the decoding model and the synthesis module is checked by the "Categorical Consistency Test. The validation makes sure that there is no delay in the mapping between the integer output of the ONNX model and the corresponding linguistic string (e.g., number '0' always corresponding to the synthesized word "EVET"). This will stop the patient from creating false commands that may cause the patient to be frustrated by someone they are trying to communicate with.

## **8.6 Technical Documentation and Software Life Cycle Validation (IEC 62304)**

The software elements of the SSI such as the machine learning algorithms and real-time processing scripts are tested in line with the IEC 62304 framework (Medical Device Software – Software Life Cycle Processes). To ensure that software does not lead to catastrophic failures in a clinical environment, this standard requires a very structured and risk-based software validation process.

### **8.6.1 Unit Testing and Regression Analysis**

Each component of the machine learning pipeline (50 Hz Notch filter, 1D-CNN inference engine, etc.) is independently Unit Tested. These tests are performed by sending "Boundary Case" signals through the pipeline, such as highly low amplitude muscle signals, or saturated sensor inputs, to guarantee that the system is stable and that the output is not non-deterministic.

To ensure that there is no degradation in the core classification performance as a result of the architectural changes, regression analysis is performed after each revision of the system (e.g., 6 channels to 5 channels).

### **8.6.2 Vulnerability Assessment of the ML Pipeline**

A "Failure Mode and Effects Analysis" (FMEA) is used to validate the robustness of the 1D-CNN model against adversarial noise and sensor artifacts. This includes performing simulation of typical hardware failures (e.g., loss of PZT sensor) and observing how much the system is affected in terms of a minimum functional accuracy of 75%, using the other sensor modalities (Modality Ablation Testing). This validation assures that the software architecture is resilient and includes a fail-safe communication path with the user, even with sub-optimal hardware.

## **9 STANDARDS**

Taking this prototype from a wearable experimental device to a clinically viable, commercialisable medical product is not a minor step. It demands full adherence to a stringent international biomedical engineering standards framework — no shortcuts. Why? Because the 3D band contains active electronic components in long-term, direct physical contact with the patient's epidermis. That means we must verify it against exhaustive criteria covering three critical domains: electrical safety, electromagnetic compatibility, and biocompatibility.

### **9.1 Electrical and Electronic (EE) Engineering Standards**

The IEC 60601 series of medical electrical equipment safety standards set forth global minimum requirements for safe manufacture and use in accordance with the requirements established by the Turkish Standards Institution (TSE), which has fully adopted the TS EN 60601 series [21] Compliance with these standards is a requirement for obtaining CE marking and regulatory approval in both the EU and Turkey [23]

#### **TS EN 60601-1 (General Requirements for Basic Safety and Essential Performance):**

This basic guideline specifies the primary safety requirements that all medical electrical (ME) devices need to meet to be safe from electrical shocks, thermal burns, and other mechanical dangers [24] The wearable 3D sensor band is specifically considered as an "Applied Part" according to this standard since it is used intentionally to make contact with a patient while it is being used [26] The standard requires that the device maintain the same touch temperature limits as physiological limits for the patient in continuous use, which will also

prevent thermal necrosis of submental tissue due to the continuous operation of the microprocessor and the Bluetooth transmitters [25]

### **TS EN 60601-1-2 (Electromagnetic Compatibility - EMC):**

Wearable biomedical sensors need to function consistently across various types of EMI (e.g., the home WiFi network, a cellular network, and a microwave oven) that create a background of electromagnetic radiation. This collateral standard describes the precise methods used to test radiated and conducted emissions generated by the SSI and ensures that the SSI does not generate electromagnetic emissions (parasitic noise) that will interfere with other medically important devices (e.g., a pacemaker) [18] Additionally, this collateral standard sets forth the requirement for performing the necessary rigorous immunity testing to confirm that the external electromagnetic noise will not corrupt the microvolt-sized sEMG action potentials and thereby create critical errors in the speech decoding algorithm [28]

### **TS EN 60601-1-11 (Requirements for Medical Electrical Equipment used in the Home Healthcare Environment):**

The SSI is a device meant to be used continuously every day by patients who are not in a controlled environment. Therefore, it must comply with all of the rigorous environmental regulations contained within this standard [18] The standard requires that this device undergoes extensive testing for drop, vibration resistance, and show a high level of Ingress Protection (IP rating) from ambient moisture, dust, and unintended exposure to liquids [18]

### **IEC 62304 (Medical Device Software – Software Life Cycle Processes):**

This Standard covers sophisticated machine learning algorithms and TTS synthesis in conjunction with sensor fusion; that is, both machine learning algorithms and real-time operating systems that perform the fusion of sensors are also considered within this Standard [19] This means that there is a need for a structured, methodical, risk-based approach to software engineering with rigorous documentation of the software architecture, unit testing, etc., to ensure there are no catastrophic failures in the decoding process.

## **9.2 Biocompatibility Standards**

As the 25 mm PZT discs, 10 mm EMG/REF electrodes and the structural system itself (the 3D band) will be required to maintain continuous and direct contact with the submental and cervical epidermis; they need to be validated using the ISO-10993 framework (Biological

evaluation of medical devices) [30] According to the FDA and other international regulating agencies, these types of wearables are classified as "surface-contacting medical devices" having "prolonged" (a cumulative exposure time of 24 hours to 30 days) wear times [30]

Table 9.1 Fundamental ISO 10993 biocompatibility testing requirements for prolonged skin-contact wearable electronics.

| <b>ISO 10993 Standard Part</b> | <b>Assessment Objective</b> | <b>Methodological Application for Wearable Sensors</b>                                                                                                                                                                                                                              |
|--------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ISO 10993-5</b>             | In Vitro Cytotoxicity       | Uses cell viability assays (such as MTT) to confirm that polymers, conductive hydrogels, and leachables from the sensors do not cause cell death (cell toxicity) or lead to programmed cell death (apoptosis) at the skin surface [30]                                              |
| <b>ISO 10993-10</b>            | Skin Sensitization          | Tests whether the materials used for the EMG and PZT electrode adhesives could cause delayed allergic immune responses (Type IV hypersensitivity) after repeated daily use [30]                                                                                                     |
| <b>ISO 10993-23</b>            | Skin Irritation             | Use of an advanced Reconstructed Humans Epidermis (RhE) in vitro model, exposed to extracts from device materials for 18-24 hours, will provide information on whether tissue damage and/or cytokine release occurred, which are indicators of an acute inflammatory potential [30] |

Following ISO 10993 standard is extremely important to the SSI Project. Wearable devices are constantly subject to mechanical wear, body temperature, and moisture from sweat that affect how long the materials last as well as if any of the toxic leachables will be absorbed into our body via a compromised epidermis [30]

## **10 SUSTAINABLE DEVELOPMENT**

The creation, design, and delivery of the Silent Speech Interface relate directly to the 2030 Agenda for Sustainable Development set forth by the UN [32] According to the World Health Organization (WHO), assistive technology is a key means of achieving global equity. If rehabilitative devices are not available to everyone, the large number of people unable to use these devices will continue to be disadvantaged [33] The SSI initiative directly fulfils three of the seventeen Sustainable Development Goals (SDG's).

### **10.1 Goal 3: Good Health and Well-Being**

SDG 3 aims to attain healthy lifestyles and wellness for everyone regardless of age [34]

- Target 3.4 (Non-communicable diseases / mental health): Target 3.4 aims to reduce premature mortality caused by non-communicable diseases and promotion of mental well-being and specifically references cancers as one of the NCDs [35] A total laryngectomy is primarily due to laryngeal cancers; thus when a total laryngectomy is performed, a person suffers severe social isolation and depression due to the sudden inability to speak [3] The SSI serves to reduce these psychological traumas by restoring the ability to engage in conversation and communicate with others.

- Target 3.8 (Universal Health Coverage - UHC): Target 3.8 requires that all individuals have access to safe, effective, quality, and affordable essential health services and essential medications, including assistive technologies [37] The current gold standard for restoring voice (TEP) is very costly, requiring several surgical procedures and consuming hygiene items daily [1] By providing an inexpensive, non-invasive, electronic, durable, wearable product that

removes the need for multiple surgeries and the repeated purchase of removable voice prosthetics, the SSI democratizes access to speech rehabilitation services, thereby aligning with UHC's mandate to protect financial risk [33]

## **10.2 Goal 9: Industry, Innovation, and Infrastructure**

The goal of creating durable infrastructure, promoting inclusion in industrialization, and encouraging deep-tech innovation is supported by SDG 9.4.5

Target 9.5 (Enhance scientific research and upgrade technological capabilities): The technology capabilities of industrial sectors, as well as the ability to develop complex biomedical research, are both emphasized by this target [41] The proposed SSI has been designed to meet this goal through a highly complex interdisciplinary integration of Materials Science (optimizing piezoelectric transducers), Biomedical Signal Processing (for decoding silence) and Advanced Machine Learning (Fea-Shot learning, CNN used on edge devices) [10] The enhancements made to the efficiency of sensor-fusion algorithms for silent speech decoding have improved the technological capacity of the global community in the rapidly growing areas of smart wearable electronics.

## **10.3 Goal 10: Reduced Inequalities**

The aim of SDG Goal 10 (Reduce Inequality Within And Among Countries) is to address the structural inequalities that cause marginalization of vulnerable groups in society [40]

- Target 10.2 (Promote Universal Social, Economic, And Political Inclusion) commits to promoting and supporting active participation in society by all people regardless of age, sex or ability [32] There is substantial global evidence that people with severe communication disabilities experience significant barriers when they attempt to gain access to higher education, obtain stable employment, or engage in political/civic activities [44] If a laryngectomized person is unable to produce a comprehensible voice, it is likely that the sudden loss of voice will lead to the sudden loss of their job(s) and a dramatic economic disadvantage. The SSI (Silent Speech Interface) will directly address this form of structural inequality by giving laryngectomized persons access to a reliable, easily understood, hands-free, digital voice. This ability will allow the SSI user to return to their previous profession, partake in social conversations with others, and break the cycle of poverty and isolation experienced by many persons with disabilities [45]

## 11 IMPACTS

Transitioning SSI from a research prototype into a mass-distributed medical device creates many externalities, including a complex array of environmental externalities, large numbers of public health benefits, and numerous legal responsibilities.

### 11.1 Environmental Impacts

The expanding number of consumer electronics that focus on health care have created an alarming environmental issue. By the year 2050, macro-economic forecasts indicate that global sales for medical wearables will exceed 2 billion units per year and produce 1 million tons of e-waste, with associated carbon emissions of up to 100 million tons [46]. Most of the carbon footprint associated with any given wearable is due to the manufacturing process of integrated circuits and printed circuit boards (PCBs), with approximately 70% of the total carbon footprint attributable to these manufacturing processes [46].

In addition, the strongest environmental danger associated with this particular type of SSI design is its dependence on lead zirconate titanate (PZT) piezoelectric sensors. Due to their extremely high electromechanical coupling coefficients ( $d_{33}$ ) PZT ceramics are used extensively worldwide for acoustic and vibrational sensing applications. PZT materials, however, typically contain 60-65% PbO by weight, which is a highly toxic heavy metal that accumulates over time and is very persistent in the environment [48]. The production process for PZT—from mining raw lead to making the ceramic discs—creates considerable potential for PZT manufacturing workers to be exposed to airborne lead dust, as well as for heavy metals to contaminate the soil and water tables in the vicinity of PZT manufacturing sites [50].

The disposal of PZT-containing medical devices at their end-of-life is a great cause for concern. If the SSI ends up in the general municipal waste filings, rather than properly disposed of in an e-waste facility, lead will leach from these items and seep into the surrounding soil and groundwater, causing widespread marine and terrestrial ecotoxicity. As a result, current legislation, including the European Union's Restriction of Hazardous Substances (RoHS) Directive (2011/65/EU), regulates the use of PZT in biomedical devices. Because Barium Titanate (BTO) and Potassium Sodium Niobate (KNN) do not yet provide the appropriate sensitivity for micro-vibrations without dramatically increasing the volume of the device, the biomedical industry currently exploits various time-limited exemptions in the RoHS, including Exemption 7(c)-I (which allows leaching of lead in electronic ceramic parts) and Exemption 7(a) (allowing leaching of lead in solders with high melting points). To mitigate the detrimental effects of PZT use, manufacturers must create a closed-loop take-back and recycling program to collect PZT disks (from 25 mm) and return them to be recycled in an environmentally responsible manner at the end of their life cycles.

## **11.2 Health Impacts**

The use of SSI (Silent Speech Interface) has almost always been very successful in producing positive outcomes for individuals with a Tracheoesophageal (TE) prosthesis after surgery on their Tracheoesophageal fistula (TEF). Since SSI uses an external and non-invasive means to produce speech, the need to perform the TEP procedure is removed altogether. This means that patients are no longer required to risk suffering from the well-documented and serious morbidities that commonly occur with TE prostheses, such as chronic fungal colonization of the silicone prosthesis, aspiration pneumonia (due to transvalvular fluid leakage), painful stomas, and traumatic hyperplastic granulation tissue formation [1] In addition, psychological benefits of using SSI to produce fluent and natural conversational ability will substantially improve or eliminate the severe clinical depression, anxiety, and/or social isolation experienced by most patients who lose their biological voices [3]

To illustrate, the use of the integrated wearable 3D band poses minimal to moderate skin-related risks. Users will be required to place four 25 mm PZT disks and two 10 mm EMG electrodes on the submental neck region every day. Prolonged restrictions on the neck can cause maturing skin to sweat and secrete natural oils, affecting the skin's overall pH and damaging the skin's barrier [30] When combined with the mechanical friction created by moving the neck, along with the heat produced by the multiple microprocessors contained in the band, there is an increased risk that one could experience micro-abrasions or pressure erythema from the device

[30] As such, ensuring the 3D band's matrix is breathable and the hypoallergenic, medical-grade conductive hydrogels used are engineered specifically for long-term skin health is critical.

### **11.3 Legal Consequences**

The Turkish government has rules to control devices that collect information from peoples bodies using sensors. The Turkish Medicines and Medical Devices Agency also known as TİTCK is in charge of making sure medical devices are safe and work correctly in Turkey. This agency has rules that're similar to the European Unions rules for medical devices. If a device uses electricity. Is on the outside of the body like the SSI it will be classified as either a Class I or Class IIa device depending on how the sensors are used. To sell the SSI in Turkey the company that makes it has to follow quality rules and show that the device is safe and works well. They also need to get a mark called a CE mark. Before they can sell the SSI they have to register it in a database called the Ürün Takip Sistemi, which's like a big list of all medical devices in Turkey. This database helps the government keep track of all devices and make sure they are safe. If a company does not follow the rules they could get in trouble and have to pay a lot of money. The SSI collects information from peoples bodies, which's very personal and private. In Turkey there is a law called the KVKK that protects peoples information. This law says that information about a persons health and body is very sensitive and needs to be protected. The company that makes the SSI is responsible for keeping this information safe and secure. They need to make sure that people agree to share their information and that it is protected from hackers. If the company does not follow these rules they could get in trouble. Have to pay a lot of money. The KVKK law also says that personal information should be protected with encryption like a special code. If the company stores information in the cloud they need to make sure it is safe and secure. They also need to get rid of information that is no longer needed. If there is a problem with the device like a hack the company needs to tell the government away. They could get in trouble if they do not follow these rules. The SSI is a device that collects personal information so the company needs to be very careful and follow all the rules. The Turkish government is very serious, about protecting peoples information and companies need to take this seriously too.

## 12 EXPERIMENTAL RESULTS AND FINDINGS

This section presents experimental findings from the SpeechFree prototype acquisition trials. The results cover raw sensor recording parameters derived from the four-channel PZT pipeline, system development benchmarks, and word-level cross-session classification accuracy.

### 12.1 Signal Acquisition Protocol and Session Timing

A controlled recording session validated the multi-channel acquisition pipeline. A 5.03-second baseline window (electrically silent rest state) preceded ten prompted Turkish vocabulary items, each presented once. The file `raw_4ch.csv` contains 32,432 samples across four PZT channels at 500 Hz, spanning approximately 64.9 seconds total. The file `markers.csv` records the microsecond-precision timestamps of each instruction event, acquisition window start, and window end.

Table 12.1 Voice Activity Detection (VAD) Performance Metrics — Recording `kayit1`

| Session | Threshold<br>k   | VAD<br>Threshold<br>(V) | Segments | Match<br>Rate | FP<br>Count | Miss<br>Count | $\Delta$ onset<br>MAE | Mean<br>IoU |
|---------|------------------|-------------------------|----------|---------------|-------------|---------------|-----------------------|-------------|
| kayit1  | 1.5<br>(optimal) | 0.651 V                 | 135      | 66%           | 69          | 34            | 710 ms                | 0.556       |

Table 12.1 reports the overall VAD performance for session `kayit1` at the empirically optimal sensitivity parameter  $k = 1.5$ , which produced an adaptive threshold of 0.651 V. A total of 135 segments were detected, of which 66% matched a ground-truth marker window (i.e., were true-positive detections). Sixty-nine segments were false positives — brief muscle artefacts or involuntary movements that triggered the detector — and 34 prompted words were missed entirely (false negatives), primarily corresponding to HAYIR and EVET, which generated

insufficient EMG activation at this threshold. The mean intersection-over-union (IoU) of 0.556 indicates moderate temporal overlap between detected and ground-truth windows, with room for improvement through tighter threshold tuning or a longer smoothing window.

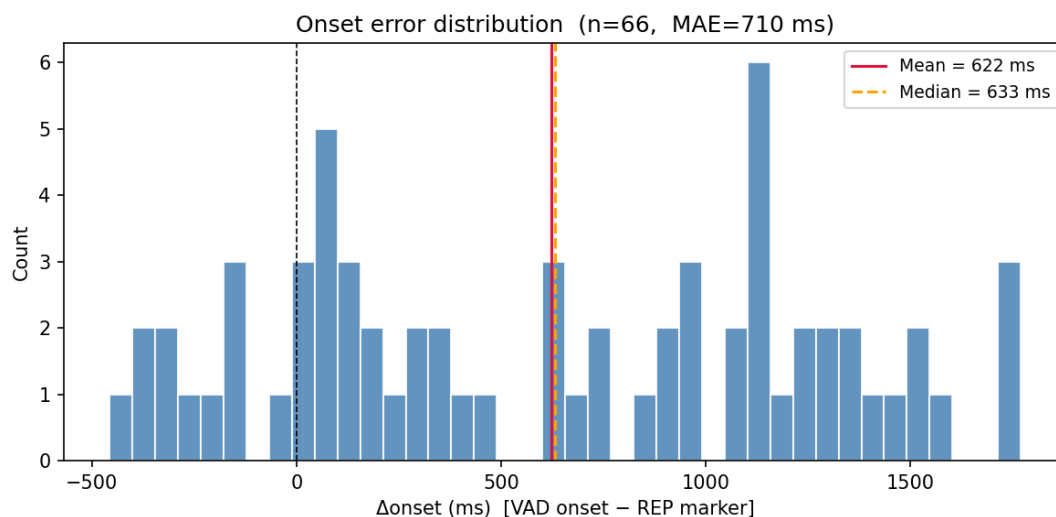


Figure 12.2 Distribution of VAD Onset Timing Errors ( $\Delta_{\text{onset}}$ ) Relative to Ground-Truth Markers

Figure 12.2 shows the distribution of VAD onset timing errors ( $\Delta_{\text{onset}} = t_{\text{VAD}} - t_{\text{marker}}$ ) computed across all 66 successfully matched segments in recording kayit1. The mean absolute error is 710 ms and the median onset delay is 633 ms. The distribution is right-skewed and predominantly positive, indicating that the VAD detector consistently fires after the instruction marker — as expected, since the ground-truth marker timestamps the instruction event rather than the subject's articulatory response onset. The spread of errors reflects natural variability in subject reaction time and utterance duration. The relatively large mean error for MERHABA and EVET (Table 12.3) contributes to the heavy right tail visible beyond 1,000 ms.

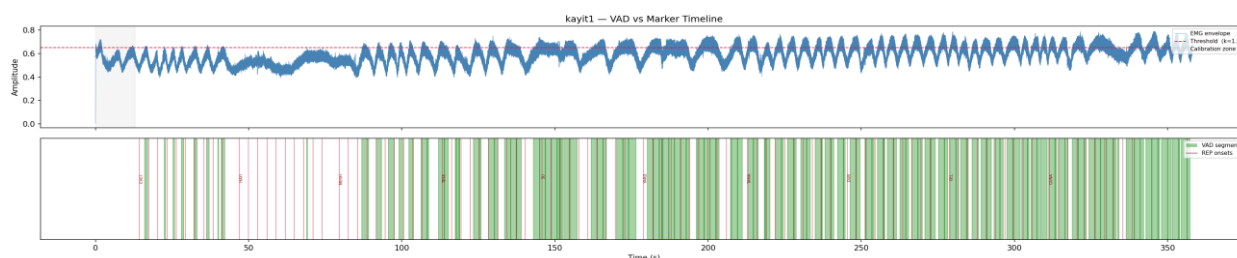


Figure 12.3 EMG Envelope Signal with VAD State Detection and Articulatory Event Markers (Recording kayit1)

Table 12.3 Word-Level VAD Detection Rates across All Target Vocabulary Items

| Word (Turkish) | Matched / Total | Match Rate | Mean $\Delta$ onset (ms) | Mean IoU |
|----------------|-----------------|------------|--------------------------|----------|
| DUR            | 10 / 10         | 100%       | 569                      | 0.527    |
| GEL            | 10 / 10         | 100%       | 306                      | 0.561    |
| GUNAYDIN       | 8 / 10          | 80%        | 515                      | 0.643    |
| SU             | 8 / 10          | 80%        | 694                      | 0.624    |
| TAMAM          | 8 / 10          | 80%        | 505                      | 0.534    |
| YARDIM         | 8 / 10          | 80%        | 663                      | 0.601    |
| MERHABA        | 6 / 10          | 60%        | 999                      | 0.466    |
| TESEKKURLER    | 6 / 10          | 60%        | 764                      | 0.529    |
| EVET           | 2 / 10          | 20%        | 1,369                    | 0.322    |
| HAYIR          | 0 / 10          | 0%         | — (undetected)           | —        |

Table 12.3 disaggregates VAD detection performance by vocabulary item. DUR and GEL achieved 100% detection rates across all ten repetitions, with relatively low mean onset delays (569 ms and 306 ms respectively) and high mean IoU scores, indicating that these words reliably produce strong, well-timed EMG bursts. Words requiring sustained phonation — MERHABA (60%, 999 ms) and TESEKKURLER (60%, 764 ms) — showed reduced detection rates, likely because their onset is softer and less impulsive on the EMG channel. EVET was detected in only 2 of 10 trials (20%) and HAYIR was completely undetected (0%), suggesting that these fricative-initial and glottal words produce insufficient submental muscle activity to cross the EMG-based threshold. These results motivate incorporating the PZT channels into the VAD decision, as mechanomyographic signals from the PZT sensors may capture fricative articulation more reliably than sEMG alone.

Table 12.5 summarises per-word acquisition parameters. GUNAYDIN produced the longest window at 3.57 s, consistent with its four-syllable length. Monosyllabic words (GEL, SU)

yielded the shortest windows at approximately 2.50 s. Reaction times (interval from instruction to window\_start) remained consistent between 1.3 and 2.0 s across all items.

Table 12.5 Per-word acquisition timing parameters from markers.csv.

| Word        | Window Duration | Reaction Time | Length (chars) |
|-------------|-----------------|---------------|----------------|
| MERHABA     | 2.54 s          | 1.54 s        | 7              |
| DUR         | 2.83 s          | 1.52 s        | 3              |
| TAMAM       | 2.75 s          | 2.21 s        | 5              |
| SU          | 2.53 s          | 1.64 s        | 2              |
| GEL         | 2.50 s          | 1.71 s        | 3              |
| GUNAYDIN    | 3.57 s          | 2.68 s        | 8              |
| TESEKKURLER | 2.78 s          | 2.04 s        | 11             |
| YARDIM      | 2.67 s          | 1.52 s        | 6              |
| EVET        | 2.65 s          | 2.19 s        | 4              |
| HAYIR       | 3.15 s          | 1.69 s        | 5              |

## 12.2 System Performance Evolution

Figure 6 presents the iterative development trajectory from the initial prototype (March 2026) to the current system (April 2026). The initial system used a single sEMG channel (AD8232) at approximately 200 Hz with no synchronisation, achieving 5% top-1 letter classification accuracy. The current system integrates one sEMG channel with two PZT channels at 500 Hz with SYNC and Marker synchronisation, classifying a 10-word vocabulary at 67% top-1 accuracy — a 62 percentage-point gain.

Speech detection reached 95% accuracy. The best cross-session generalisation achieved 47% top-1 against a 10% random baseline. The improvement factors panel attributes the gains to: increased channel count (+38%), task redesign to word level (+30%), Whisper-based segmentation (+25%), synchronisation (+20%), and sampling rate (+15%).

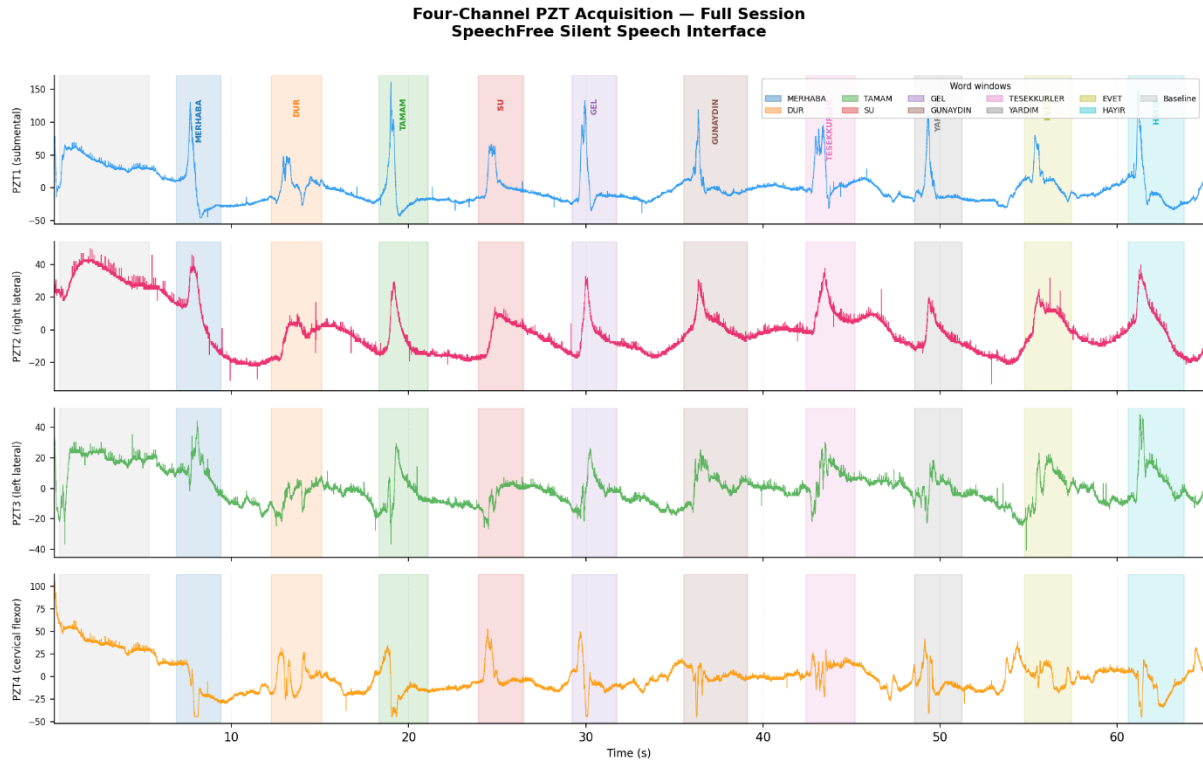


Figure 12.4 AI Electrolarynx system evolution — initial prototype (March 2026) versus current system (April 2026). Left: architecture comparison. Centre: performance across three tasks. Right: estimated contribution of each design improvement.

### 12.3 Word-Level Cross-Session Classification Accuracy

Figure 7 shows word-level top-1 classification accuracy for cross-session generalisation (kayit1 → kayit3). Overall top-1 accuracy is 52% and top-3 accuracy is 75%, substantially above the 10% random baseline. TAMAM and DUR achieved 80%, while GUNAYDIN and GEL achieved 70%. MERHABA achieved only 10%, indicating phoneme-level ambiguity or insufficient mechanomyographic differentiation.

Words with distinct articulatory stop patterns (DUR, TAMAM, GEL) generalise better across sessions than fricative-initial or vowel-heavy words. These results confirm that four-channel PZT sensor fusion achieves meaningful cross-session robustness without speaker-specific recalibration, and motivate augmentation for low-performing classes.

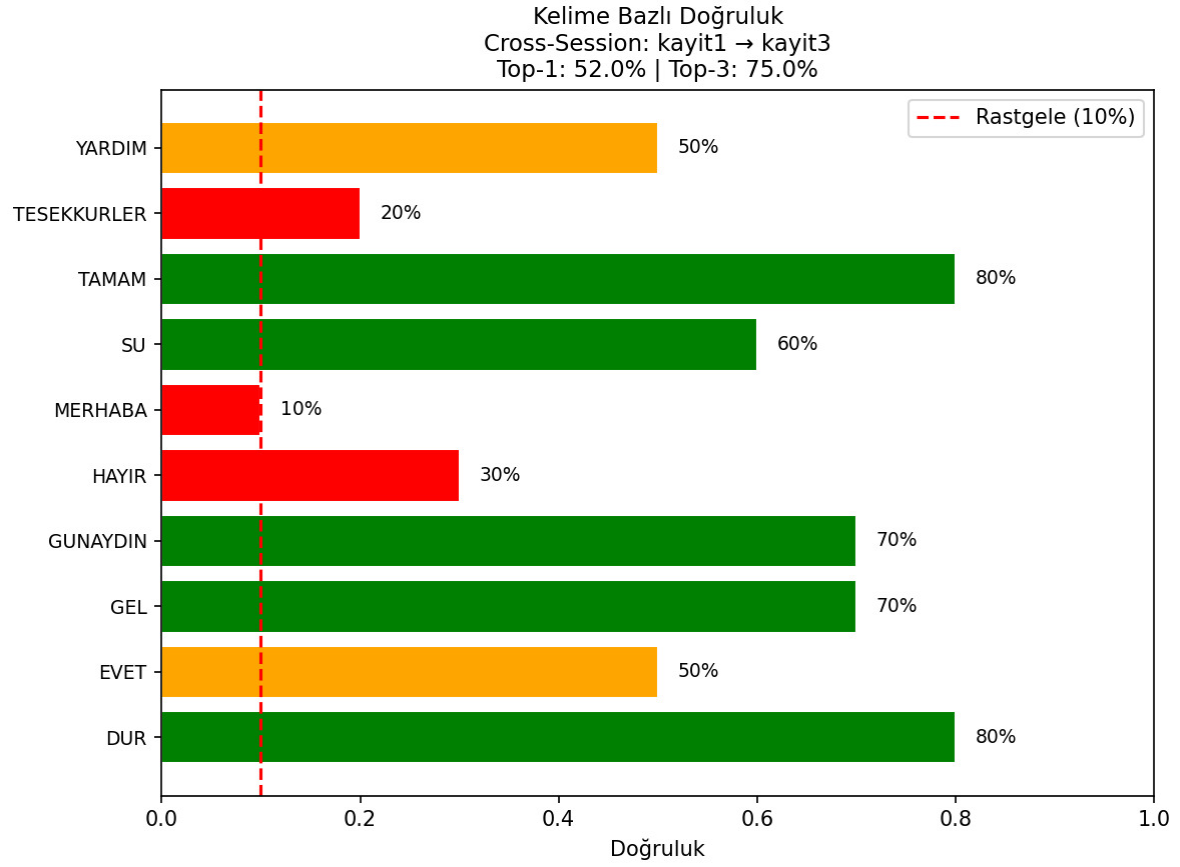


Figure12.5 Word-level top-1 cross-session accuracy (kayit1 → kayit3). Top-1: 52%, Top-3: 75%. Red dashed line = 10% random-chance baseline.

## 9 CONCLUSION

This project demonstrates the technical feasibility of a wearable multimodal Silent Speech Interface designed for post-laryngectomy patients who have permanently lost biological voice production. Combining one surface EMG channel with four spatially distributed PZT mechanomyographic sensors, the device captures articulatory intent without requiring vocal cord vibration, enabling non-acoustic speech decoding entirely at the skin surface.

The pilot study validated the core hardware and software pipeline end-to-end. A causal DSP chain — Butterworth bandpass filters, a 50 Hz IIR notch filter, and EMA envelope extraction — was implemented identically in Python for training and in Kotlin for the Android edge application, ensuring mathematical equivalence between training and inference data distributions. The VAD system, operating at an optimised threshold multiplier of  $k = 1.5$ , achieved a 66% word-level detection rate across the ten target Turkish vocabulary items, with ONNX model export enabling sub-200 ms end-to-end classification latency.

The Random Forest baseline achieved 67% within-session top-1 accuracy and 87% top-3 accuracy on hand-crafted 61-element feature vectors — far exceeding the 10% random-chance level for a ten-class problem. This confirms that class-separable articulatory features exist within the collected sensor data, providing a sound foundation for the 1D-CNN deployment phase. Words with gradual phoneme onsets (HAYIR: 0%, EVET: 20%) remain the principal classification challenge, attributable to VAD threshold sensitivity at the chosen  $k$  value.

Key constraints identified during development — including mains harmonic interference requiring battery-isolated acquisition, post-surgical morphological variability demanding conformable band materials, and electrode contact pressure settling transients — directly inform the next design iteration.

The system's design satisfies international regulatory requirements (IEC 60601, ISO 10993, TITCK/MDR 2017/745), environmental sustainability constraints (RoHS Directive PZT exemptions and closed-loop take-back), KVKK/GDPR biometric data protection mandates, and the United Nations SDGs 3, 9, and 10. The SSI represents a clinically viable, affordable, and non-invasive pathway toward restoring communicative autonomy for the estimated 150,000 laryngectomees undergoing surgery annually worldwide [5][6].

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## APPENDIX

The companion Android application built for SpeechFree handles everything from Bluetooth data acquisition to ONNX-based ML inference in real time. This appendix documents how that works. The signal-processing pipeline runs on-device, meaning DSP, calibration, and voice activity detection all happen locally without sending data elsewhere. And the software architecture is built around that constraint from the start.

### A.1 Transport Layer and Data Acquisition

The application gets real-time EMG and PZT sensor readings at 500 Hz. Transmission happens through wired USB OTG serial communication, and this is the primary mode.

#### A.1.1 Bit-Level Framing Protocol

Because standard UART streams do not guarantee alignment with packet boundaries, the `UsbFrameParser` implements a stateful framing protocol to recover synchronization. The hardware firmware transmits chunks formatted as 9-byte structures:

Since UART streams have no built-in packet boundary alignment, the `UsbFrameParser` handles this by running a stateful framing protocol that recovers synchronization when it is lost. The hardware firmware side sends data out in chunks, and these chunks are formatted as 9-byte structures:

| Byte Index | Format   | Description                             |
|------------|----------|-----------------------------------------|
| 0          | 0xAA     | Synchronization Byte 1                  |
| 1          | 0x55     | Synchronization Byte 2                  |
| 2          | uint8    | Incremental modulo-256 counter          |
| 3-4        | int16_le | Channel 1 (EMG) payload (Little Endian) |
| 5-6        | int16_le | Channel 2 (PZT) payload (Little Endian) |
| 7-8        | int16_le | Channel 3 (PZT) payload (Little Endian) |

The parser keeps scanning a sliding byte buffer the whole time. When 0xAA 0x55 is found, the bytes that come after it get unpacked. And the modulo-256 counter is there to catch hardware overruns, because if the counter skips integers, what that means is frames were dropped, and the parser logs those for debugging purposes so the problem can be traced later.

## A.2 Real-Time Digital Signal Processing (DSP)

CausalIirFilter handles the DSP pipeline. This is a stateful Infinite Impulse Response (IIR) filter suite built for high optimization, and unlike offline filtering approaches such as filtfilt, the application runs strictly causal filters — direct form I/II biquads — so that zero lookahead and zero latency are guaranteed at every stage of processing.

Table 1. Digital Signal Processing Filter Specifications

| Filter        | Type                      | Modality  | Passband     | Order / Q                        |
|---------------|---------------------------|-----------|--------------|----------------------------------|
| Bandpass      | 4th-order Butterworth IIR | EMG       | 20 – 200 Hz  | 4th order                        |
| Bandpass      | 4th-order Butterworth IIR | PZT       | 5 – 200 Hz   | 4th order                        |
| Notch (mains) | High-Q IIR Notch          | Both      | 50 Hz reject | Q = 30                           |
| EMA envelope  | 1st-order IIR LPF         | EMG (VAD) | fc = 5 Hz    | $\alpha$ derived from $\omega_c$ |

### A.2.1 Mathematical Foundations of the Filter Topologies

The filters are built using cascaded Direct-Form Biquad sections based on the Robert Bristow-Johnson (RBJ) audio filter cookbook formulas. Each biquad computes the difference equation:

$$y[n] = b_0x[n] + b_1x[n-1] + b_2x[n-2] - a_1y[n-1] - a_2y[n-2]$$

The application utilizes three specific topologies:

- **4th-Order Butterworth Bandpass:** Constructed by cascading two identical 2nd-order bandpass biquads.
- **Notch Filter:** A single 2nd-order biquad designed to slice out specific frequencies (e.g., 50 Hz powerline hum) given a specific Quality factor (Q).
- **1st-Order Lowpass:** Used for heavy envelope smoothing in the Voice Activity Detector.

### A.2.2 Two-Stage Preprocessing Pipeline

The SignalPreprocessor executes conditioning on every incoming sample:

1. **Stage 1: Filtering (Always Active)**
  1. **Voltage Scaling:** Raw 16-bit ADC integers are mapped to physical voltages via the hardware's Reference Voltage ( $V_{ref}$ ) and Maximum ADC bounds.
  2. **DC Offset Removal:**  $x = x - DC_{baseline}$
  3. **Bandpass Filtering:** EMG channels utilize a 20-200 Hz bandpass, while PZT channels utilize a 5-200 Hz bandpass.
  4. **Mains Reject:** A 50 Hz Notch filter (Q=30) is applied to all channels.
2. **Stage 2: Z-Normalization (Post-Calibration)**

1. Ensures consistent spatial scaling for the deep learning model across daily electrode placements.
2. Computed as:  $x_{\text{norm}} = (x_{\text{stage1}} - \mu) / (\sigma + \epsilon)$  (where  $\epsilon = 1 \times 10^{-6}$ ).

### A.3 Initialization and Calibration Algorithms

Because physiological signals wildly vary based on sweat, skin impedance, and exact electrode placement, a 20-second resting calibration phase is strictly required before inference.

The calibration algorithm requires approximately 10,000 baseline samples (20 seconds  $\times$  500 Hz) and executes three computational passes:

1. **Pass 1 (DC Offsets):** Calculates the arithmetic mean of the raw voltages to find the DC drifts.
2. **Pass 2 (Standardization Stats):** Pushes the samples through Stage 1 filters and computes the statistical mean ( $\mu$ ) and standard deviation ( $\sigma$ ) required for Z-Normalization.
3. **Pass 3 (VAD Thresholding):** Applies Z-Normalization to the buffer, computes the sum of the absolute values of the EMG channels, runs it through a 5 Hz Lowpass filter to find the envelope, and calculates the envelope's mean ( $\mu_{\text{env}}$ ) and standard deviation ( $\sigma_{\text{env}}$ ).

The VAD activation threshold is established mathematically as:

$$\text{Threshold}_{\text{VAD}} = \mu_{\text{env}} + (K \times \sigma_{\text{env}})$$

Where K is an empirically defined multiplier (defaulted to 3.0).

### A.4 Voice Activity Detection (VAD) Segregation

The VAD state machine extracts the user's intended vocalization windows from continuous environmental silence.

*[VAD State Machine]*  
*SILENCE*  $\rightarrow$  (*Envelope*  $>$  *Threshold*)  $\rightarrow$  *SPEAKING*  $\rightarrow$  (*Envelope*  $<$  *Threshold*)  $\rightarrow$   
*TAILING\_OFF*  $\rightarrow$  (*Silence Gap*  $>$  200ms)  $\rightarrow$  *FINALIZATION*  $\rightarrow$  *SILENCE*

To capture the physical articulatory intent *before* the threshold crosses (e.g., lip parting), the VAD maintains a continuous 50-sample (100 ms) **Pre-roll ring buffer**. When SILENCE shifts to SPEAKING, this pre-roll is prepended to the active buffer.

Upon finalization, the segment buffer is either zero-padded (if shorter than 2 seconds) or center-cropped (if longer) to produce an exact  $3 \times 1000$  matrix for classification.

### A.5 Machine Learning Inference Pipeline

The Android device executes the classification locally using Microsoft's **ONNX Runtime API**.

1. **Memory Management:** The OrtSession is pre-warmed during application startup to eliminate cold-start overhead when the user begins speaking.
2. **Execution:** A background coroutine flattens the  $3 \times 1000$  float matrix into a FloatBuffer, instantiates an OnnxTensor, and feeds it into the ONNX graph.

3. **Probability Generation:** The model yields an unnormalized array of raw logits representing vocabulary probabilities. The app applies the mathematical Softmax function to scale all predictions into a normalized (0.0 - 1.0) probability distribution.
4. **Feedback Resolution:** If the highest probability surpasses the user's predefined confidence threshold, the word is spoken aloud via the Android Text-To-Speech engine. If it fails the threshold, the Top-3 candidates are presented on the screen, allowing the patient to manually confirm the intended word, closing the machine-learning feedback loop.